PUBLIC PORTFOLIO EDITION

VEEAM BACKUP & RECOVERY LAB

This co-authored lab report documents a controlled Windows backup and file-recovery workflow.

The workstation hostname and temporary laboratory administrator account have been permanently redacted from text and screenshots. Local test paths are retained because they describe the methodology and do not identify a public target.

The original evidence remains private.

Prepared by Ammar Aikan and Aiban Mohammed

Sanitized for professional portfolio publication

VEEAM BACKUP & REPLICATION

Practical Backup and File Recovery Lab Report

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Edition
Veeam Backup & Replication Community Edition

Lab date
16 September 2026

Backup configuration • Windows Agent policy • File-level protection • Verified restore

Executive Summary

Lab Overview

This report documents a complete Veeam backup-and-recovery workflow performed on a Windows workstation. The lab created a dedicated backup repository on drive D:, configured a Windows Agent file-level backup policy for a controlled source folder on drive C:, executed and verified the backup, deliberately deleted a test file, and restored that file from the Veeam restore point. **The final verification confirmed both file presence and file content after recovery.**

BACKUP	RECOVERY	VERIFICATION
Full backup created and stored on D:	RestoreMe.txt recovered from Veeam Backup Browser	Recovered content matched the original test data

Key Outcome

The exercise achieved the intended objective: a selected folder on C: was protected using Veeam, the backup was written to the configured repository on D:, and a deleted file was successfully restored to its original location.

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1 Objective and Scope

What the lab was designed to prove

1.1 Objective

The objective of this lab was to demonstrate an end-to-end backup and recovery process using Veeam Backup & Replication Community Edition on a Windows workstation. The lab focused on a small, controlled data set so that the backup, deletion, and restore sequence could be clearly documented and validated.

1.2 Scope

- Install and launch Veeam Backup & Replication Community Edition.
- Create a Windows backup repository on drive D:.
- Prepare a controlled source directory on drive C: containing test files.
- Create a Windows Agent policy using File level backup.
- Select C:\Veeam_Lab_Source as the protected object.
- Store backups in D-Backup-Repository located at D:\Veeam_Backup_Lab.
- Run the backup manually and verify creation of Veeam backup files.
- Delete RestoreMe.txt and recover it using File Level Restore.
- Verify that the restored file is present and that its content is correct.

1.3 Success Criteria

The lab is considered successful only if the backup completes without data-loss errors, a restore point is available, RestoreMe.txt can be found inside the backup, and the restored file returns to the original source folder with the expected text content.

2 Lab Environment and Backup Design

Configuration used during the practical exercise

Item	Configuration
Backup software	Veeam Backup & Replication Community Edition
Observed build	12.3.2.3617
Protected computer	[redacted]
Agent policy type	Windows Agent Policy - Workstation
Management mode	Managed by agent
Backup mode	File level backup
Source directory	C:\Veeam_Lab_Source
Test files	File1.txt, Important.txt, RestoreMe.txt
Repository name	D-Backup-Repository
Repository path	D:\Veeam_Backup_Lab
Retention policy	7 days
Execution	Manual / no automatic schedule

2.1 Backup Flow

SOURCE	POLICY	REPOSITORY	RECOVERY
C:\Veeam_Lab_Source	File-level backup	D-Backup-Repository	RestoreMe.txt

Design note: The lab stores the backup on drive D: for demonstration. In production, a repository should be on independent storage and ideally follow the 3-2-1 backup principle so that a single physical disk failure does not affect both source and backup data.

3 Implementation and Evidence

Step-by-step configuration with screenshot evidence

3.1 Installation and Initial Preparation

The Veeam installer was launched and the full Veeam Backup & Replication product was selected for installation.

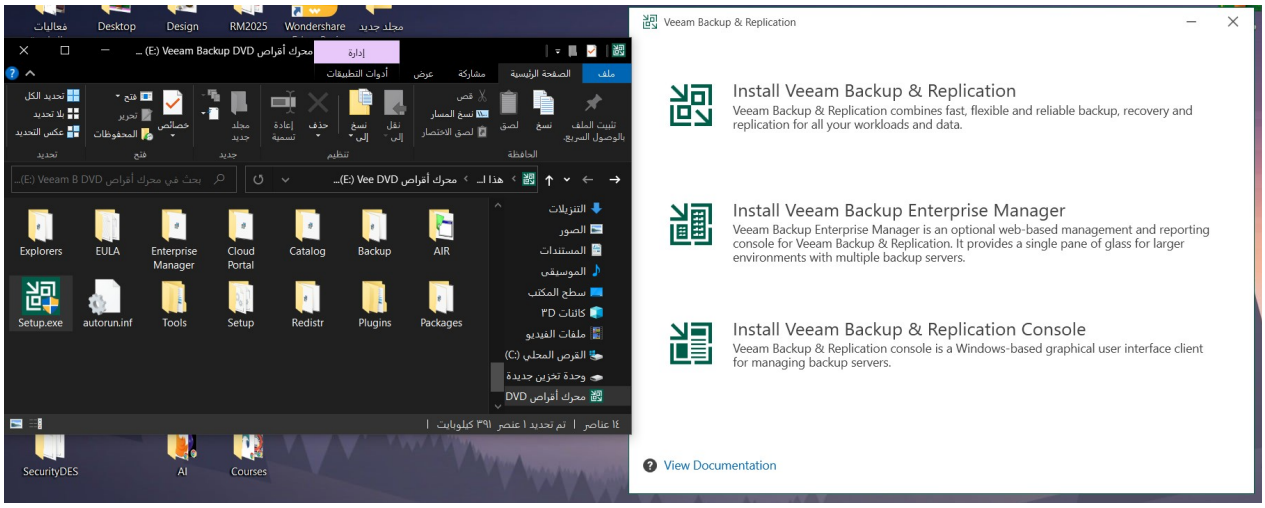


Figure 1. *Installation and product selection*

The installer displayed the final installation configuration before setup was started.

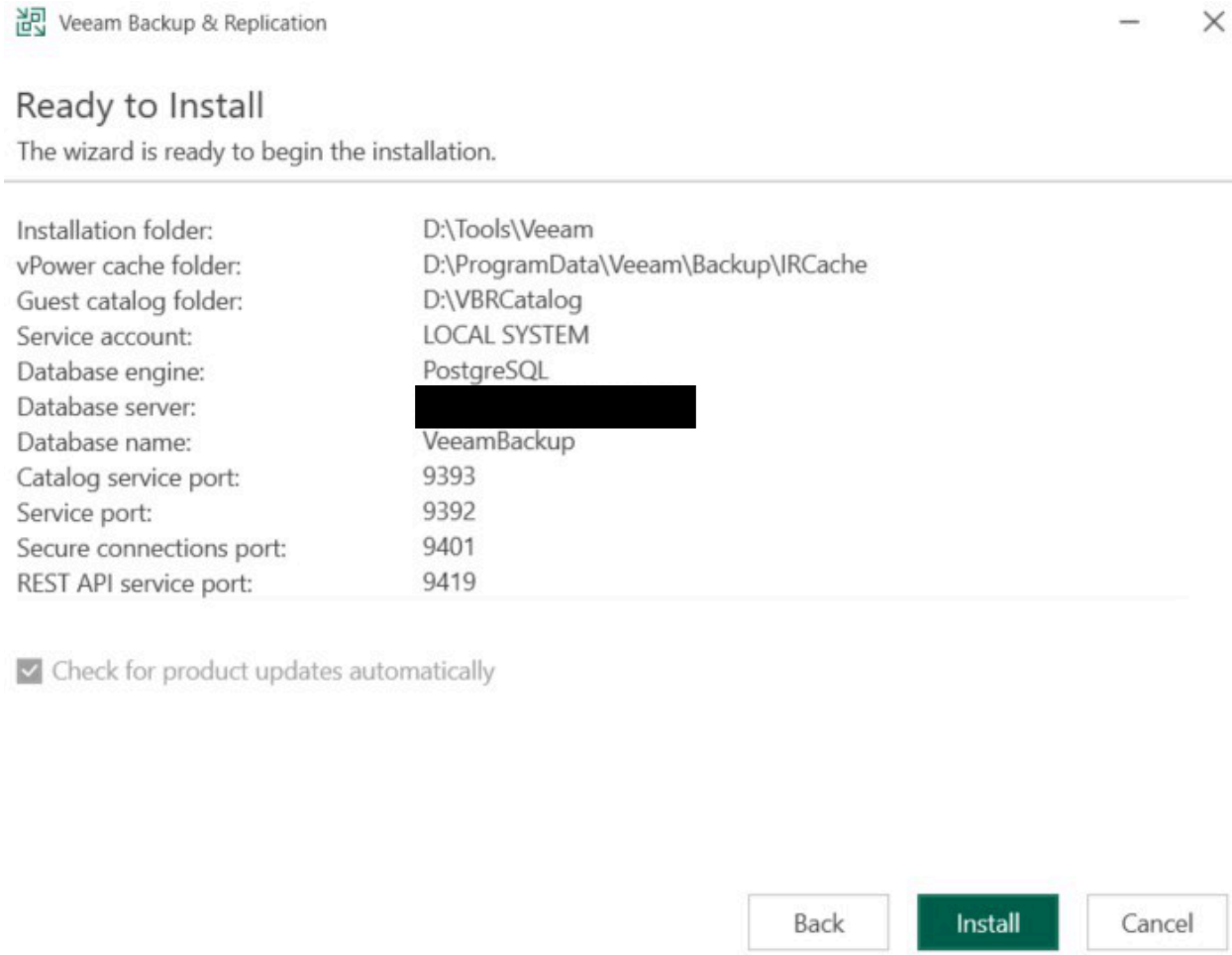


Figure 2. *Installation settings ready*

The Veeam Backup & Replication Community Edition console launched successfully.

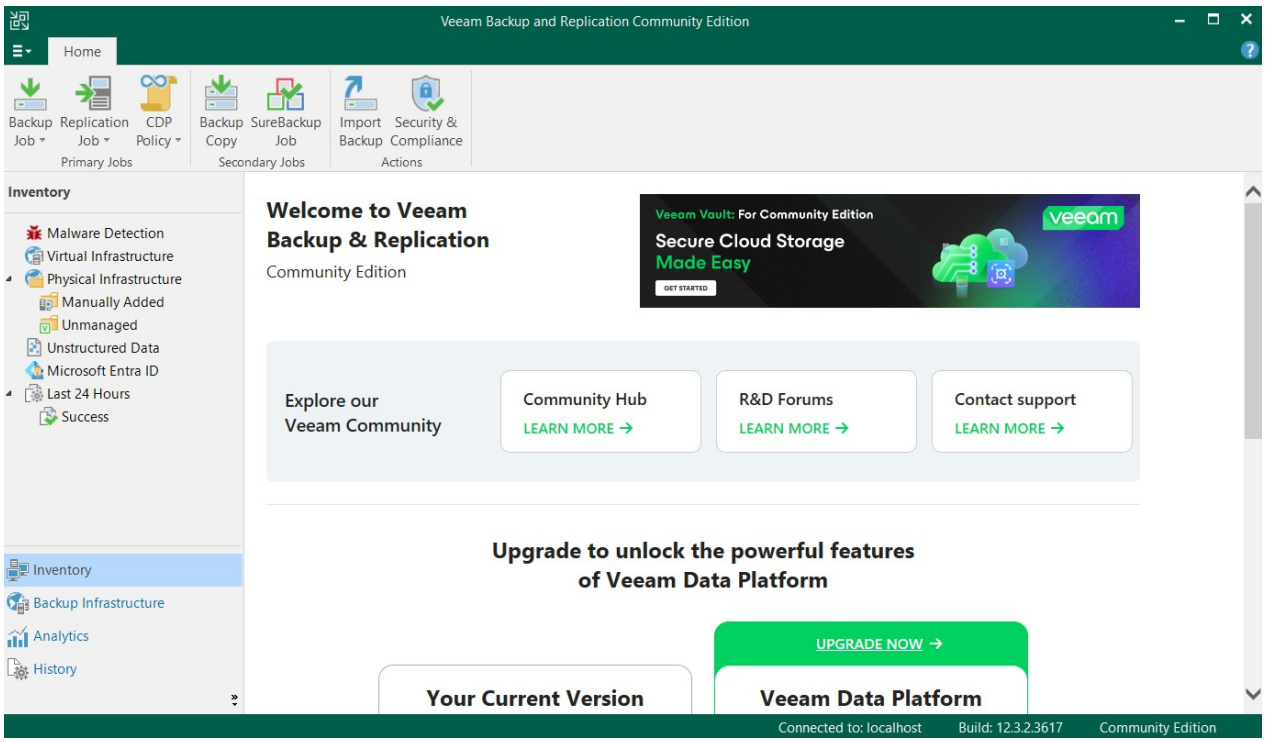


Figure 3. *Veeam Community Edition running*

PowerShell was used to create the source folder and test files on C:, and to create the destination folder on D:.

```
PS C:\Windows\system32> New-Item -ItemType Directory -Path "C:\Veeam_Lab_Source" -Force


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         9/16/2026   9:22 AM                Veeam_Lab_Source


PS C:\Windows\system32>
PS C:\Windows\system32> "Veeam Backup Lab - File 1" | Set-Content "C:\Veeam_Lab_Source\File1.txt"
PS C:\Windows\system32> "Important university backup test" | Set-Content "C:\Veeam_Lab_Source\Important.txt"
PS C:\Windows\system32> "Restore test file" | Set-Content "C:\Veeam_Lab_Source\RestoreMe.txt"
PS C:\Windows\system32>
PS C:\Windows\system32> New-Item -ItemType Directory -Path "D:\Veeam_Backup_Lab" -Force


    Directory: D:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         9/16/2026   9:22 AM                Veeam_Backup_Lab


PS C:\Windows\system32> Get-ChildItem "C:\Veeam_Lab_Source"


    Directory: C:\Veeam_Lab_Source


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/16/2026   9:22 AM             27 File1.txt
-a----         9/16/2026   9:22 AM             34 Important.txt
-a----         9/16/2026   9:22 AM             19 RestoreMe.txt


PS C:\Windows\system32> Get-Item "D:\Veeam_Backup_Lab"


    Directory: D:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         9/16/2026   9:22 AM                Veeam_Backup_Lab


PS C:\Windows\system32>
```

Figure 4. *Source and destination lab folders prepared*

3.2 Backup Repository Configuration

The existing repository configuration was reviewed before adding the dedicated lab repository.

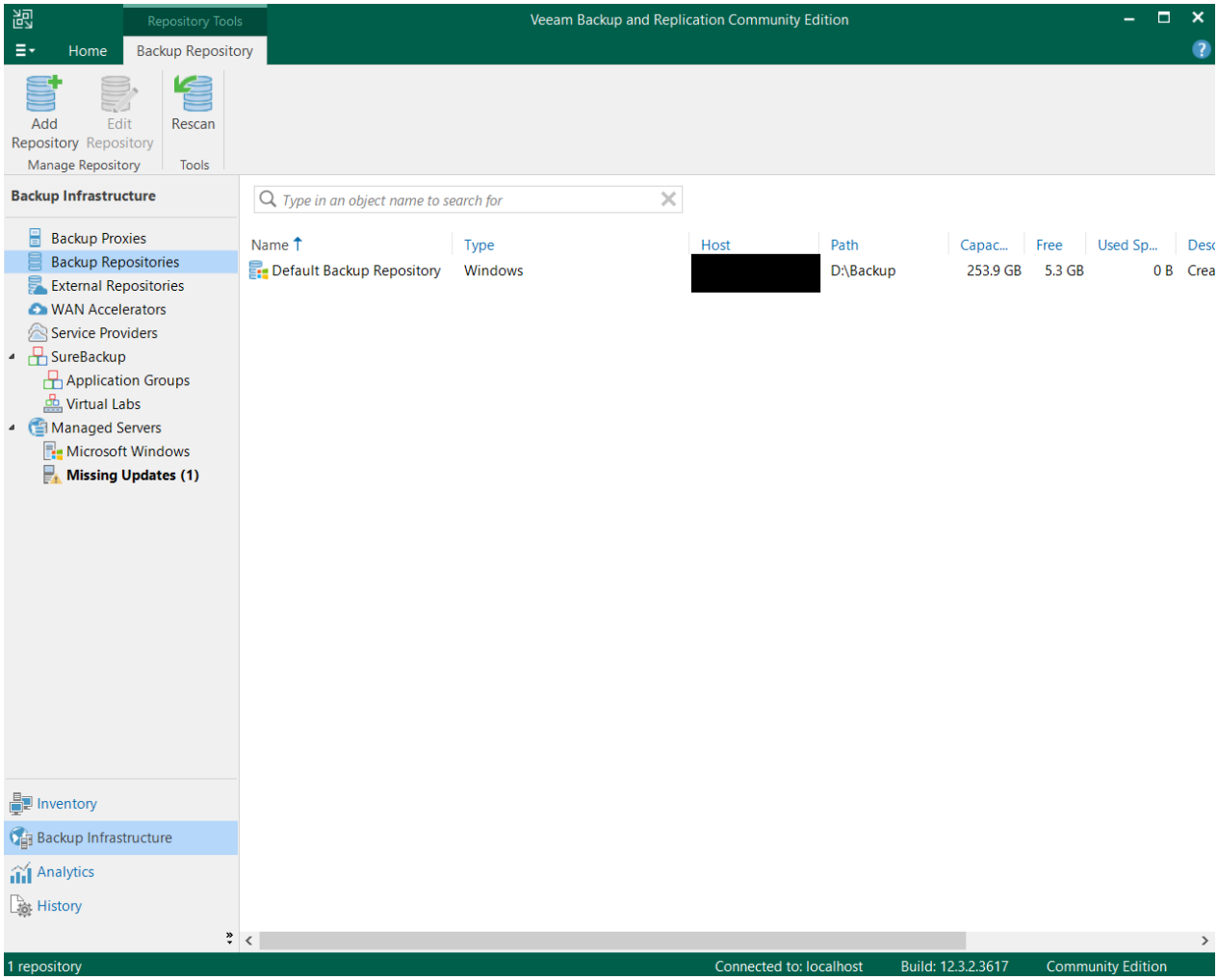


Figure 5. *Backup Repositories view before configuration*

A new direct-attached Windows repository was added for the practical backup workflow.

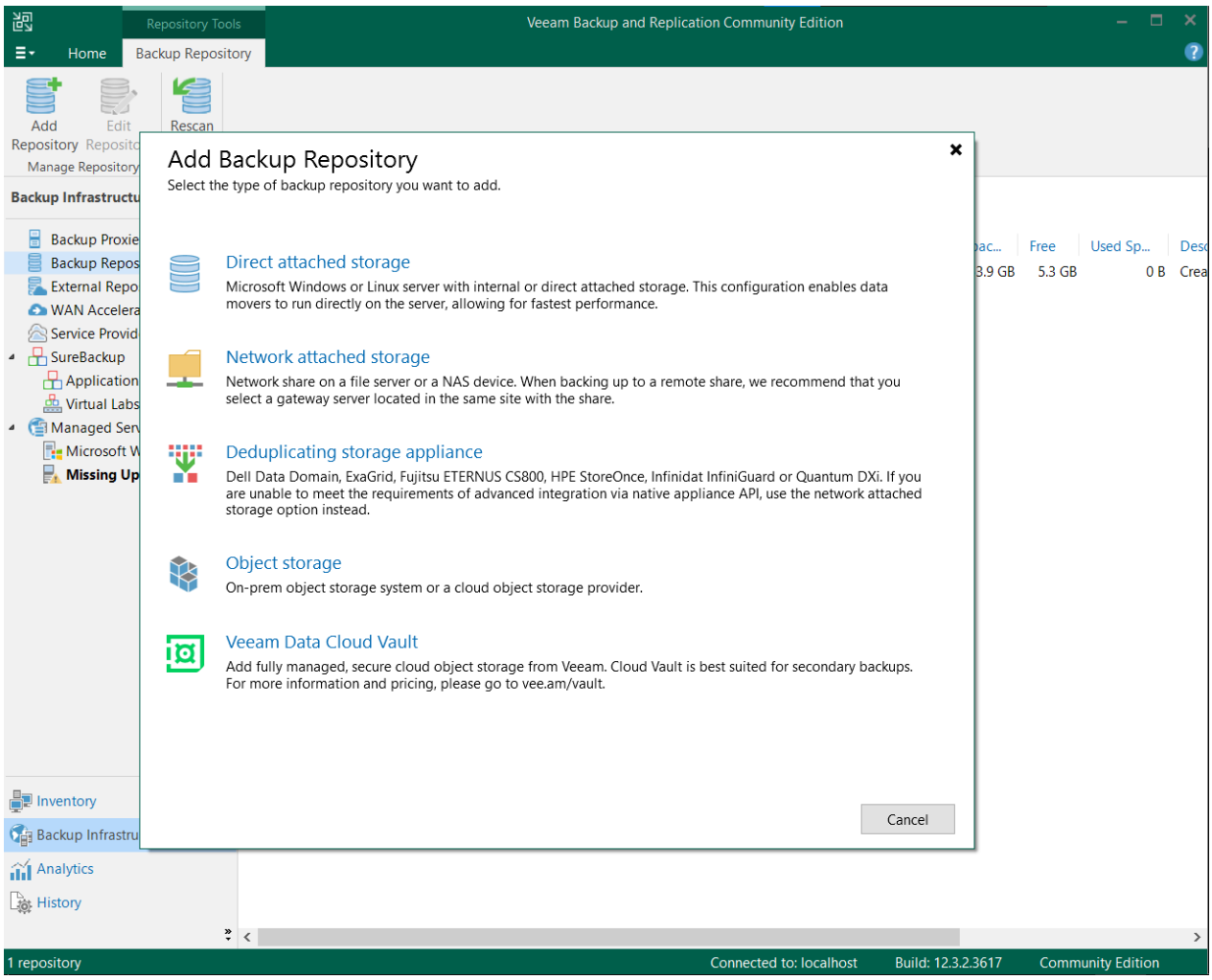


Figure 6. *Add Backup Repository wizard*

The repository storage path was set to D:\Veeam_Backup_Lab.

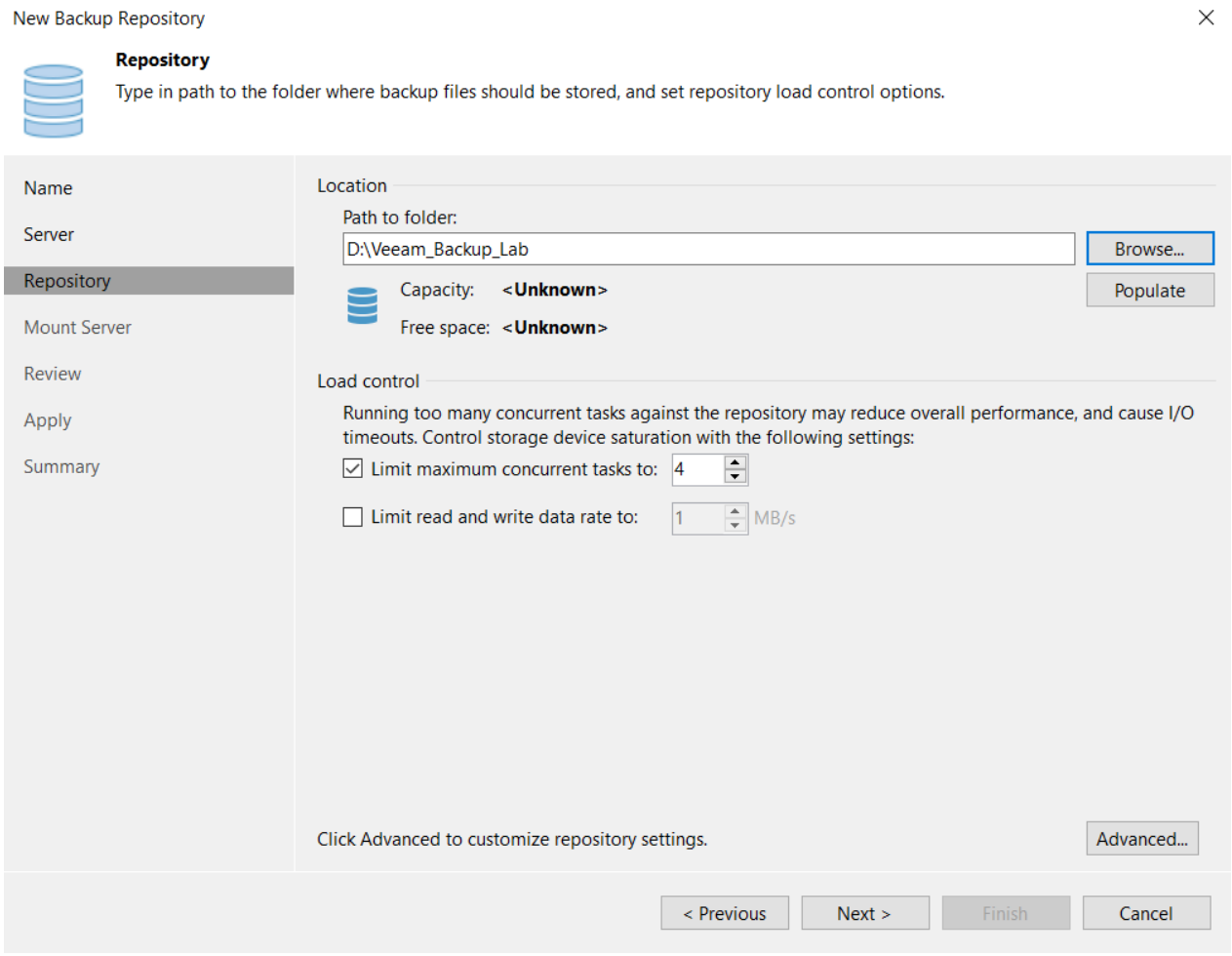


Figure 7. *Repository path configured*

The repository settings and required Veeam components were reviewed before applying the configuration.

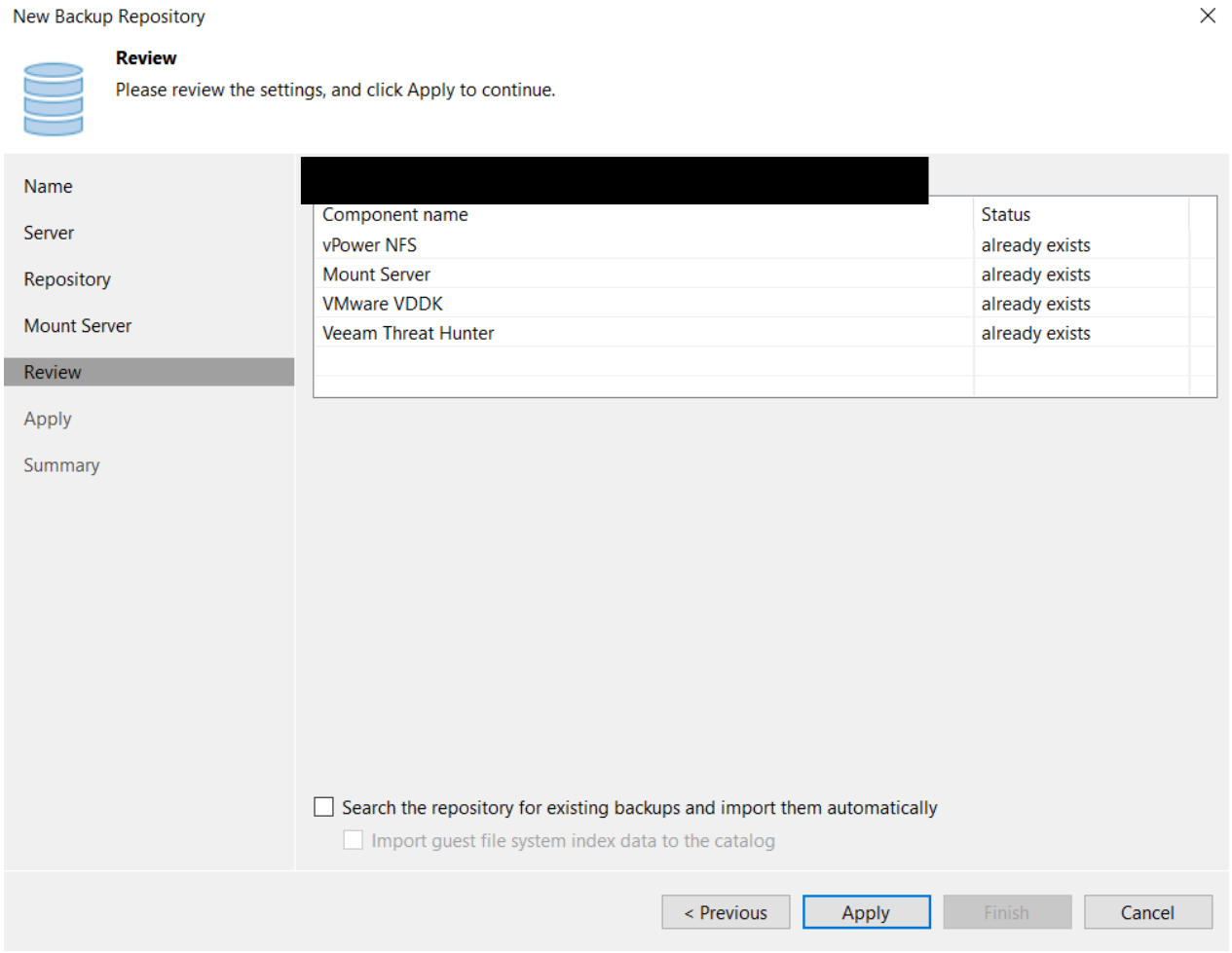


Figure 8. *Repository configuration reviewed*

Veeam confirmed that D-Backup-Repository was successfully saved and mapped to D:\Veeam_Backup_Lab.

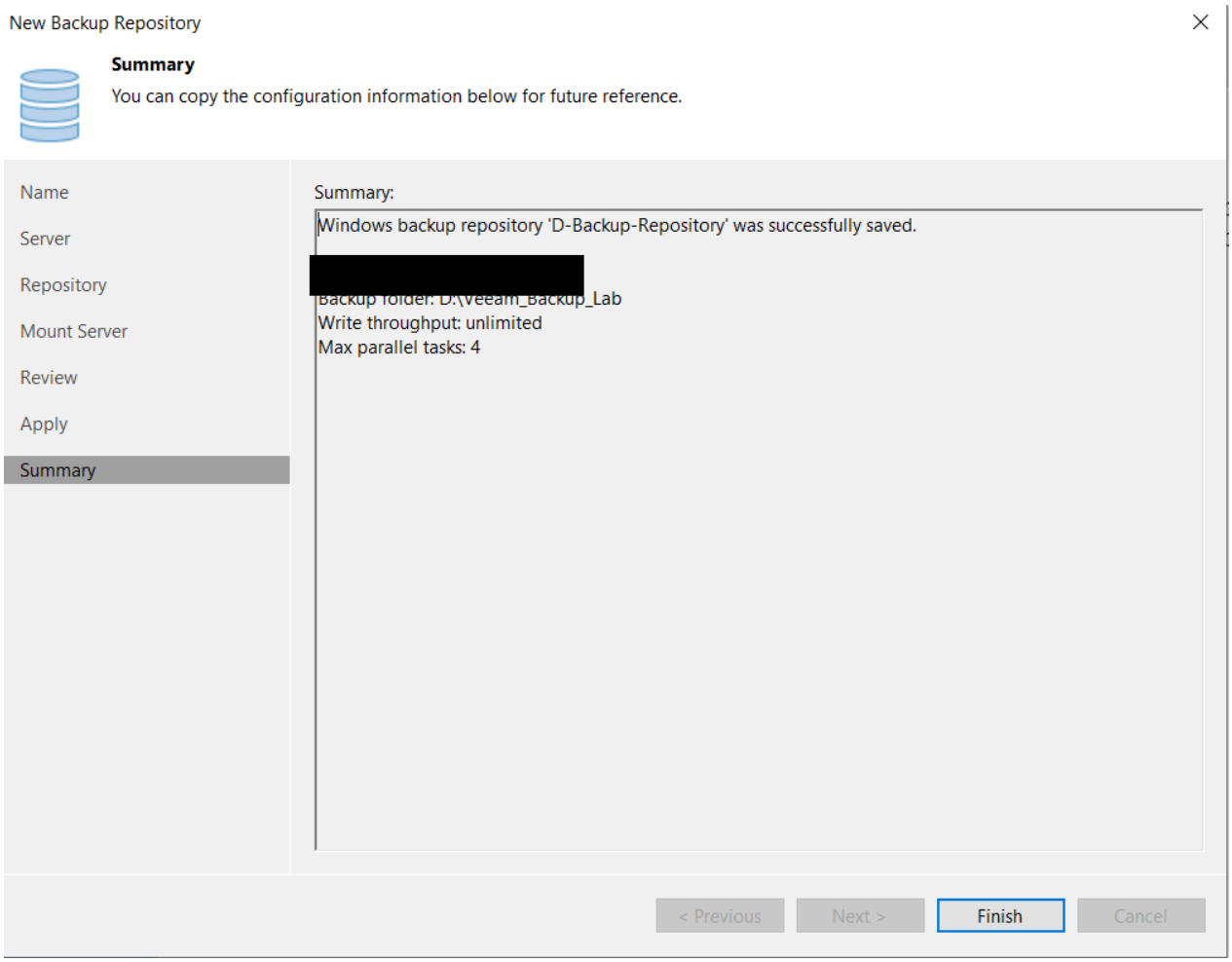


Figure 9. *Repository created successfully*

D-Backup-Repository appeared in the Backup Repositories view and was ready for use by the backup policy.

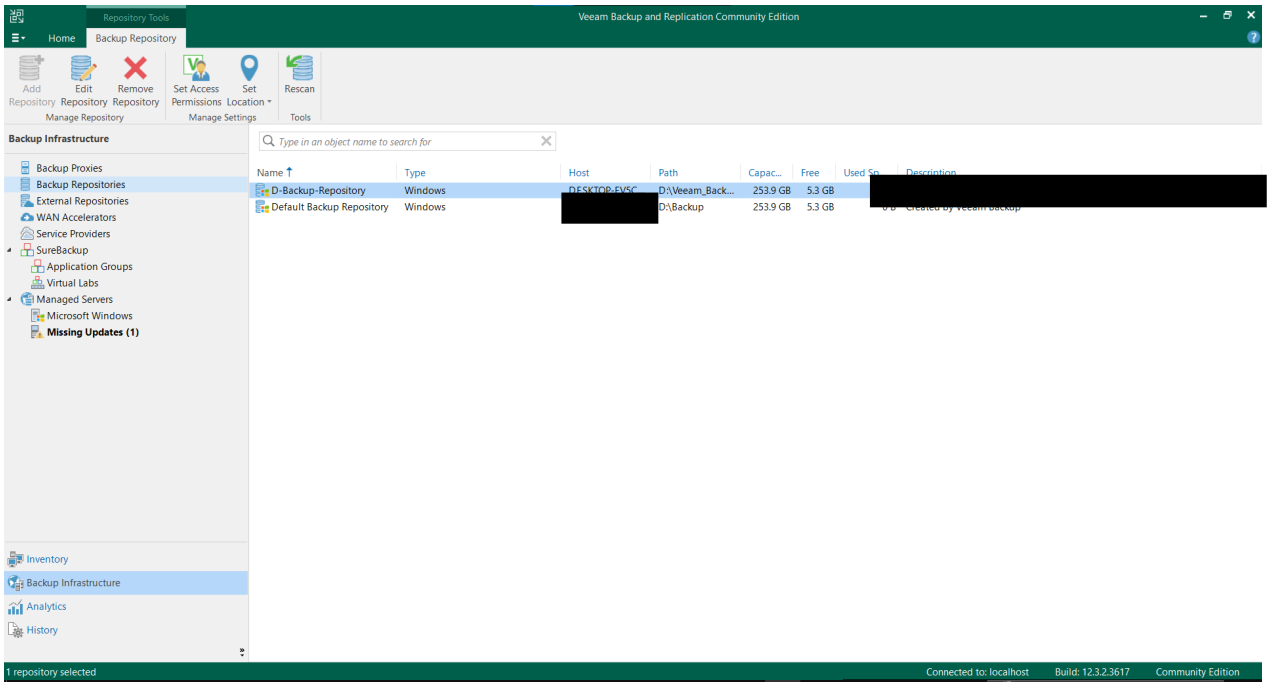


Figure 10. *Dedicated repository visible in Veeam*

3.3 Windows Agent Backup Job Configuration

A Windows computer backup job was started to protect the workstation through Veeam Agent.

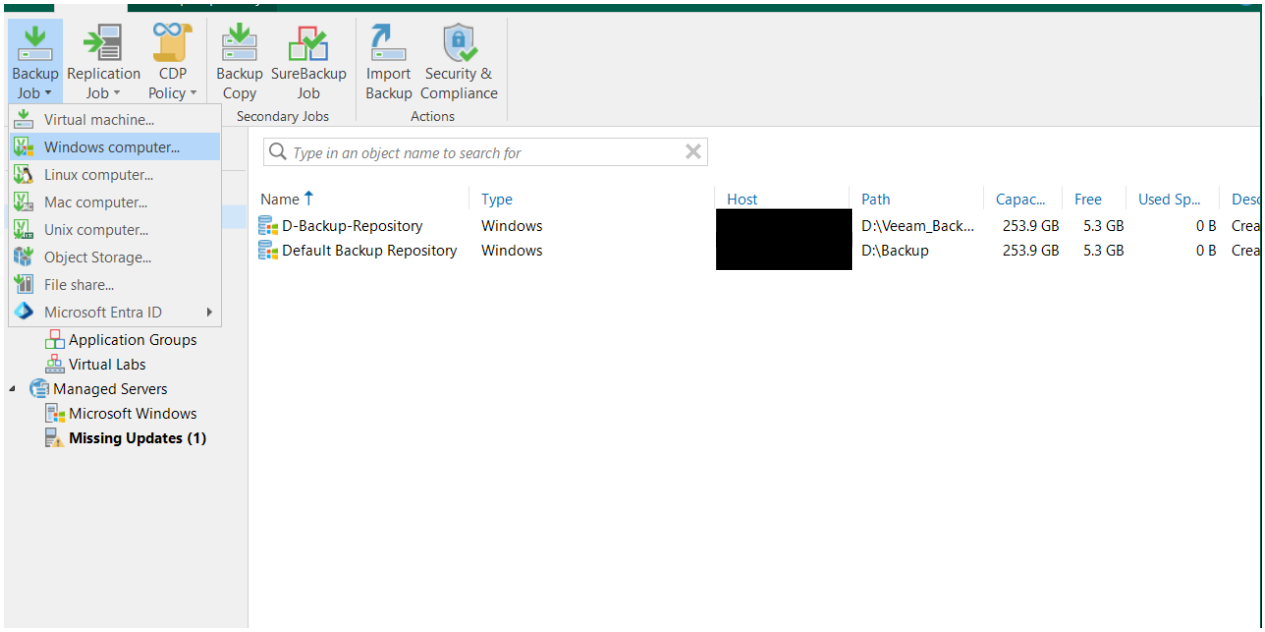


Figure 11. *Windows computer backup job selected*

The protected computer type was configured as Workstation and the policy was managed by the agent.

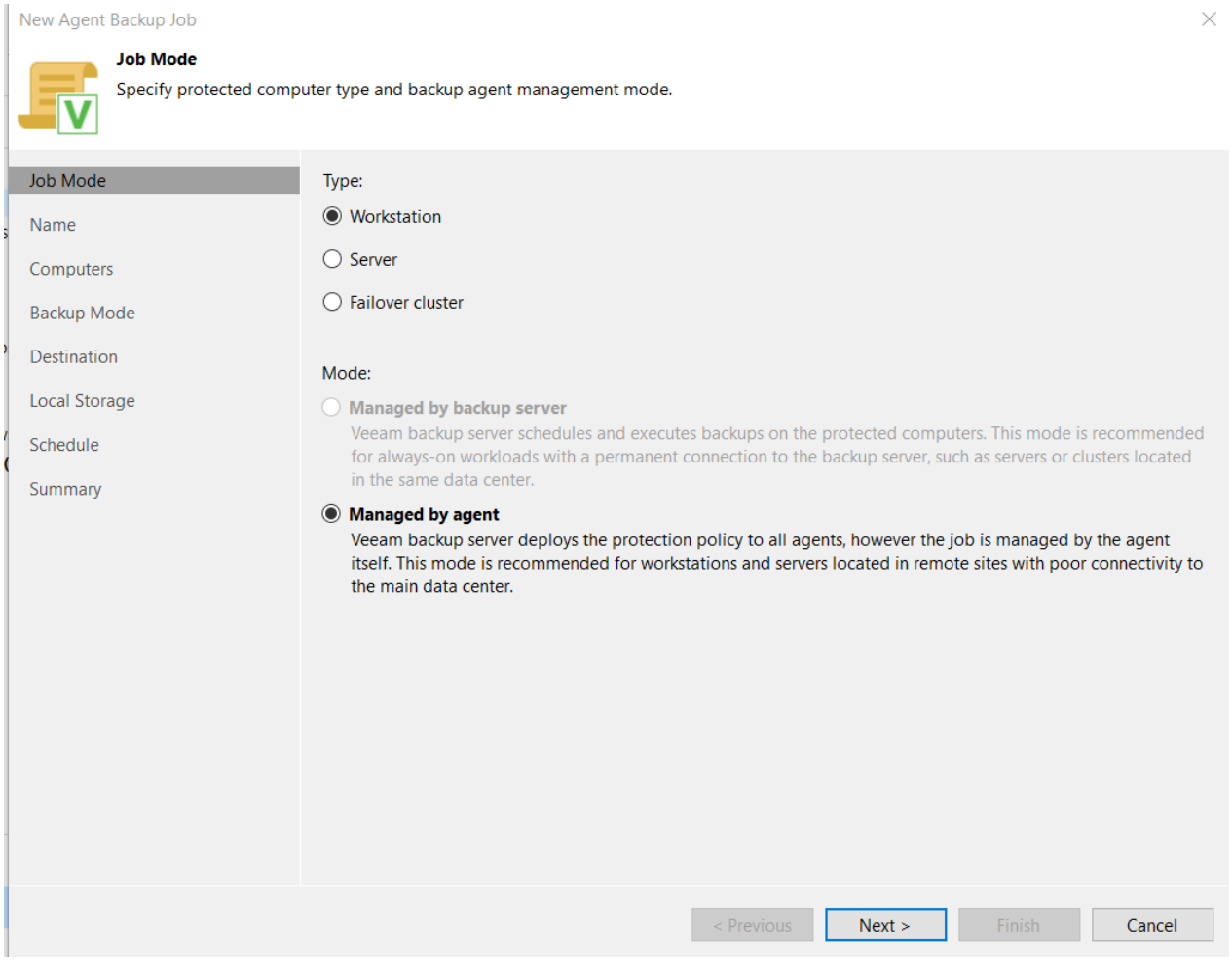


Figure 12. *Workstation agent policy selected*

The job was named C_to_D_File_Backup and its source-to-destination purpose was recorded in the description.

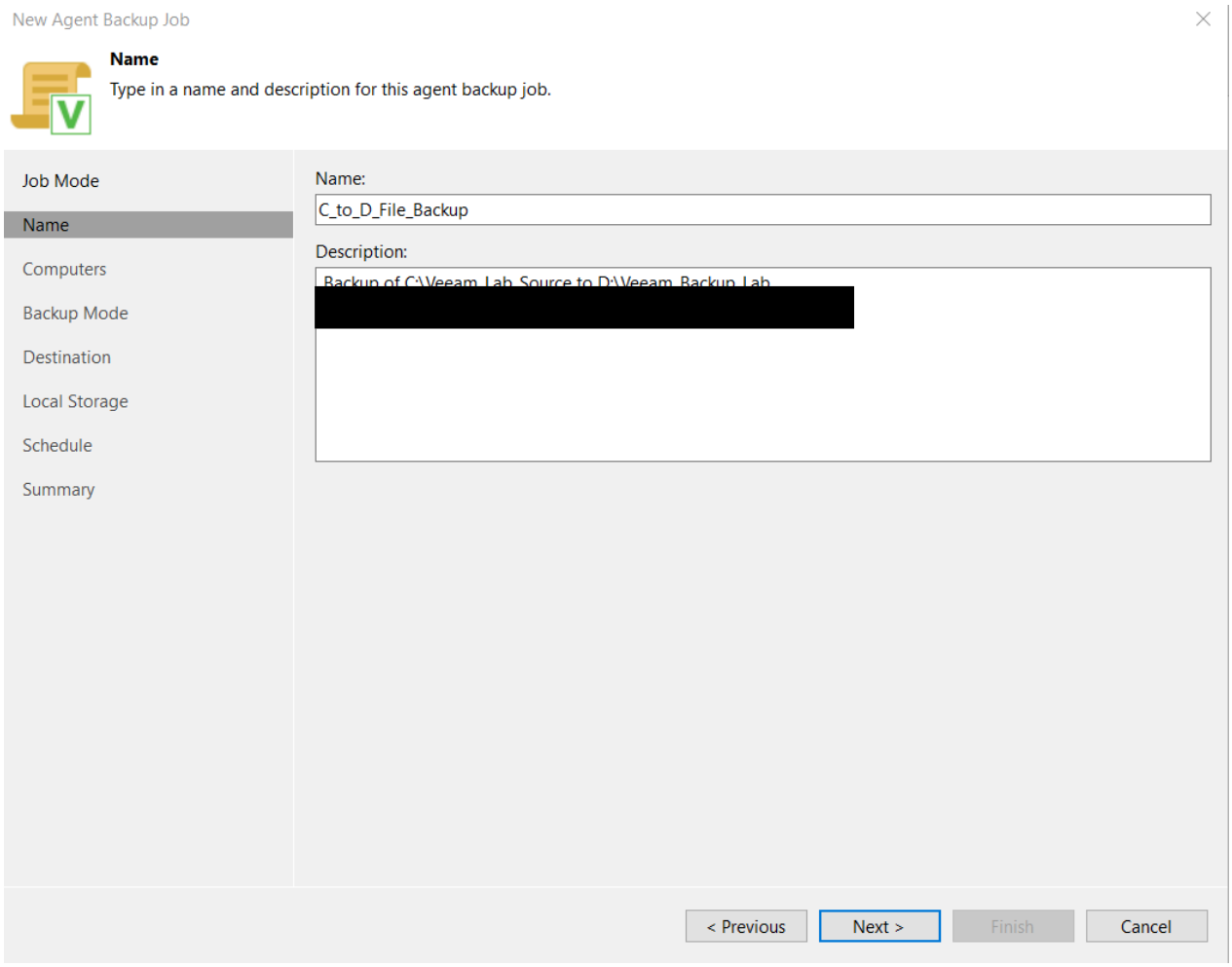


Figure 13. *Backup job named and documented*

[redacted]

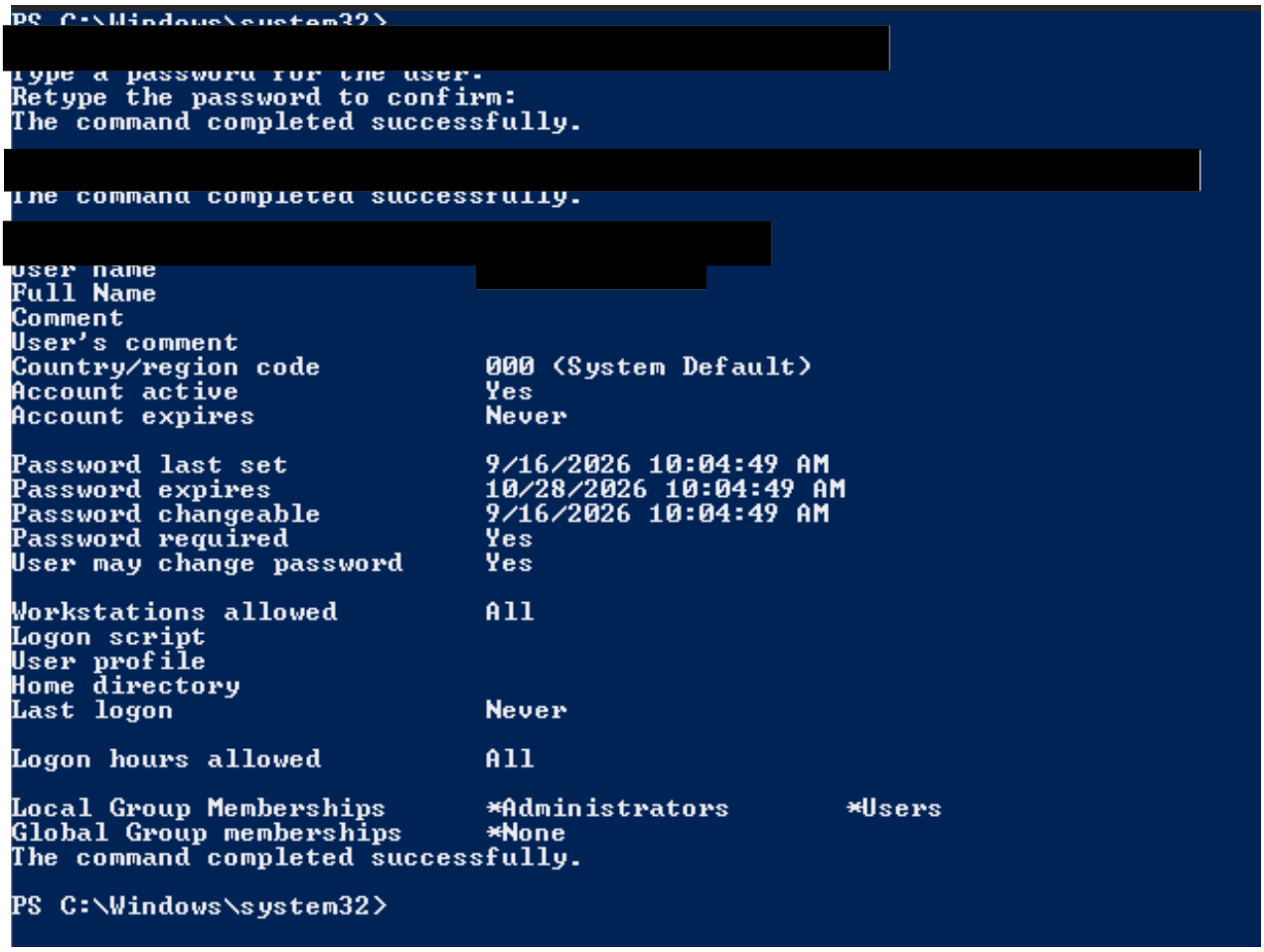


Figure 14. *Temporary local administrator created*

The temporary administrative credentials were selected for Veeam to manage the protected computer.

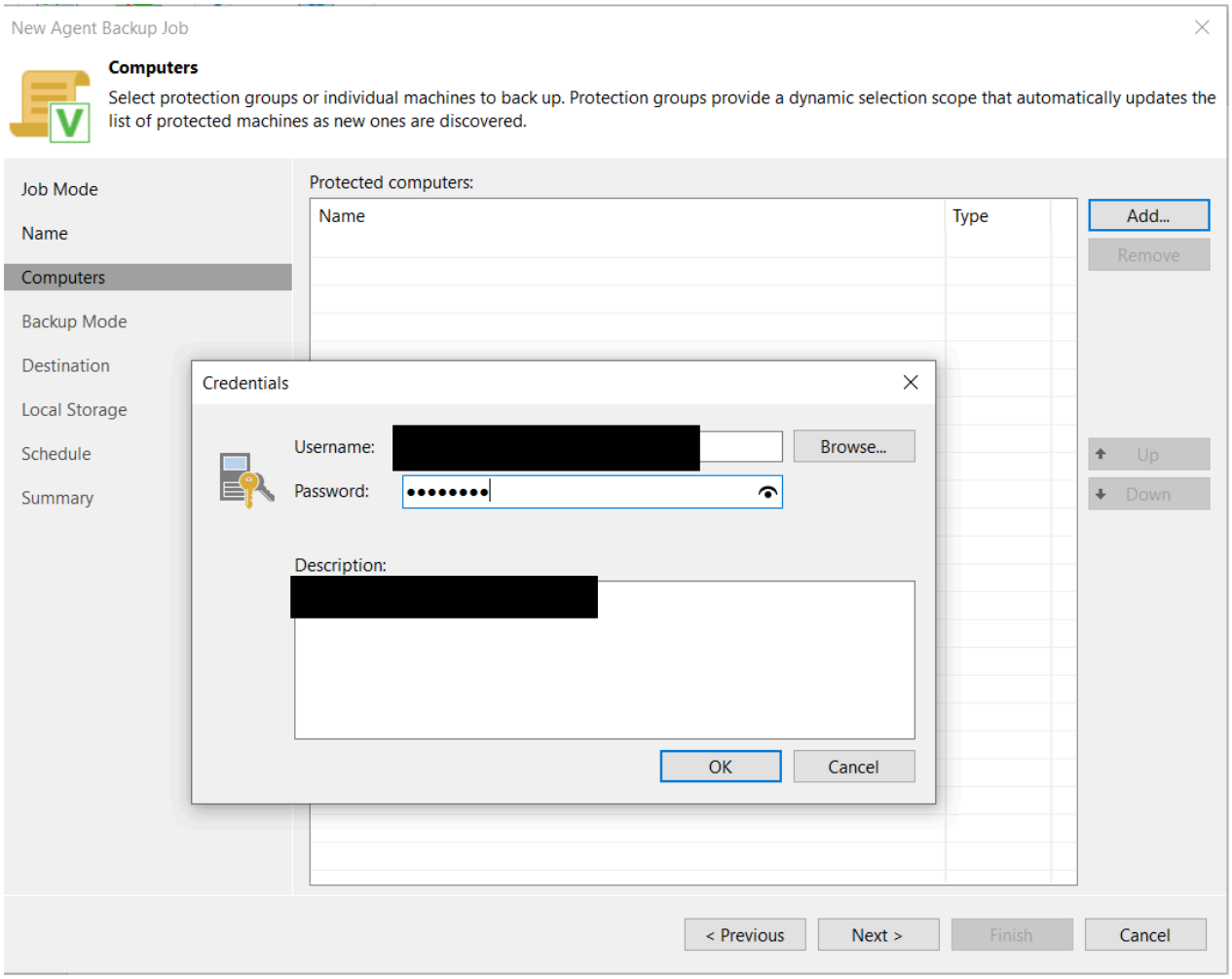


Figure 15. *Veeam credentials supplied*

File level backup was chosen because the objective was to protect a specific directory rather than the full computer or an entire volume.

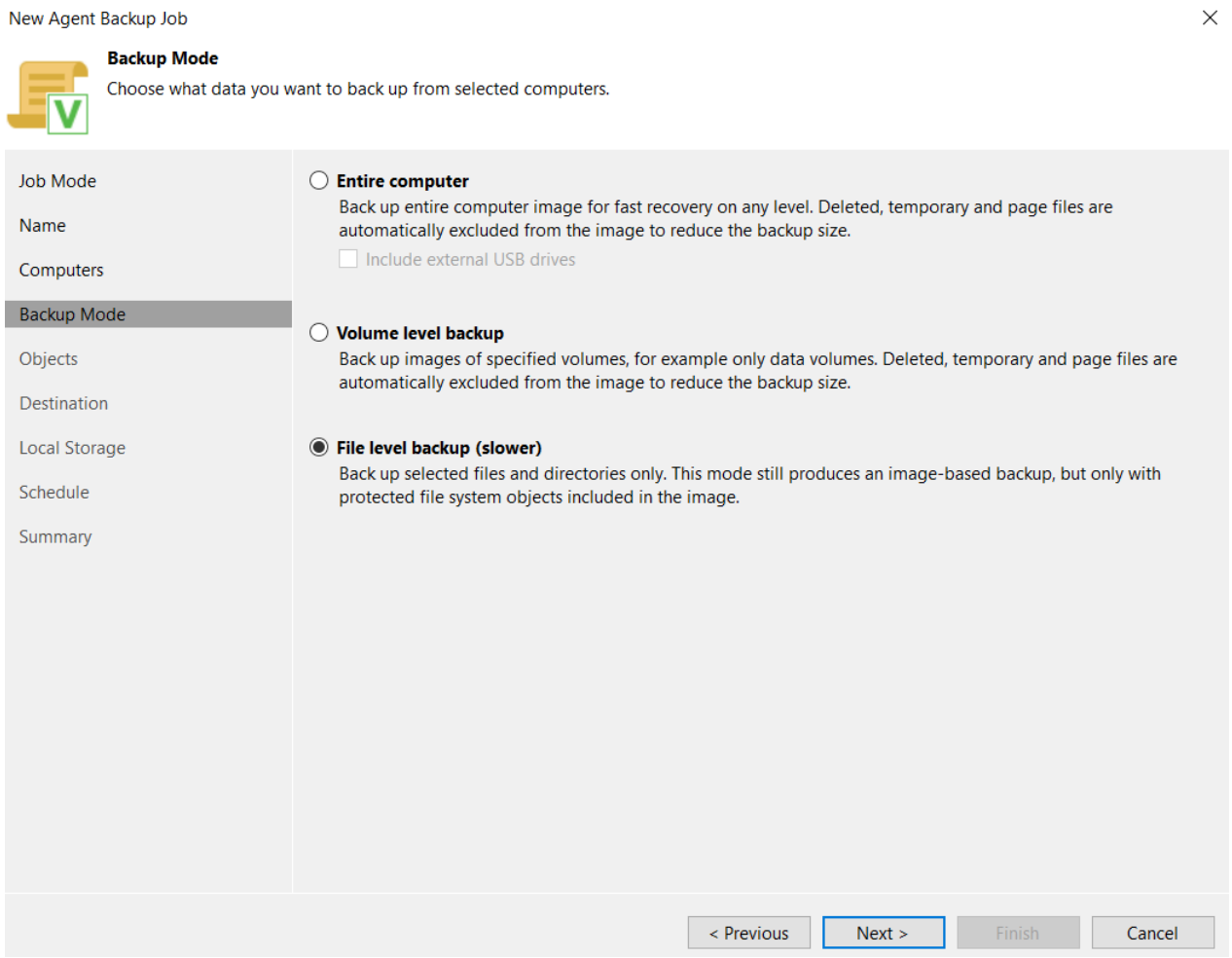


Figure 16. *File-level backup mode selected*

Only C:\Veeam_Lab_Source was selected as the file system object to protect.

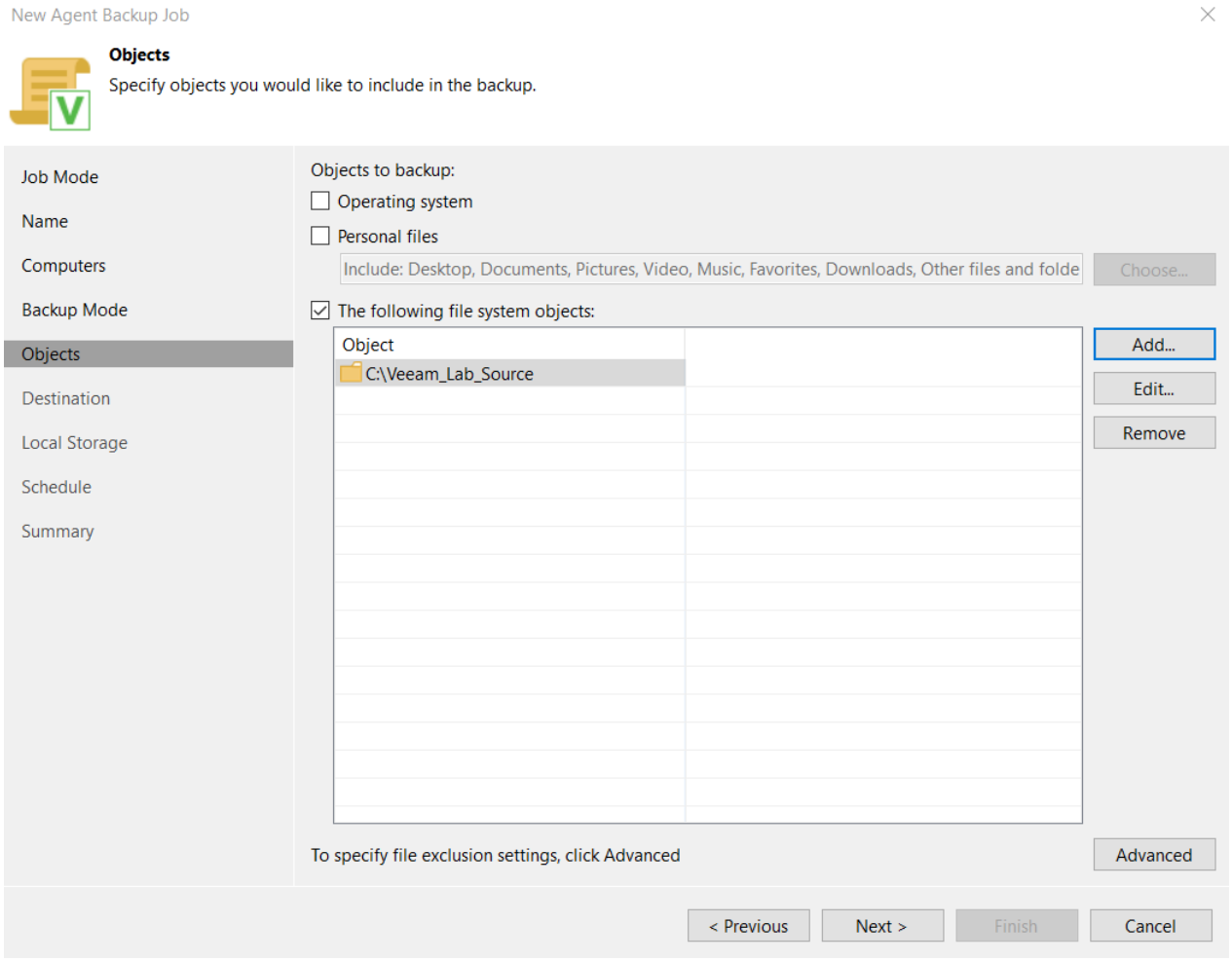


Figure 17. *Source object selected*

The job destination was set to a repository managed by Veeam Backup & Replication.

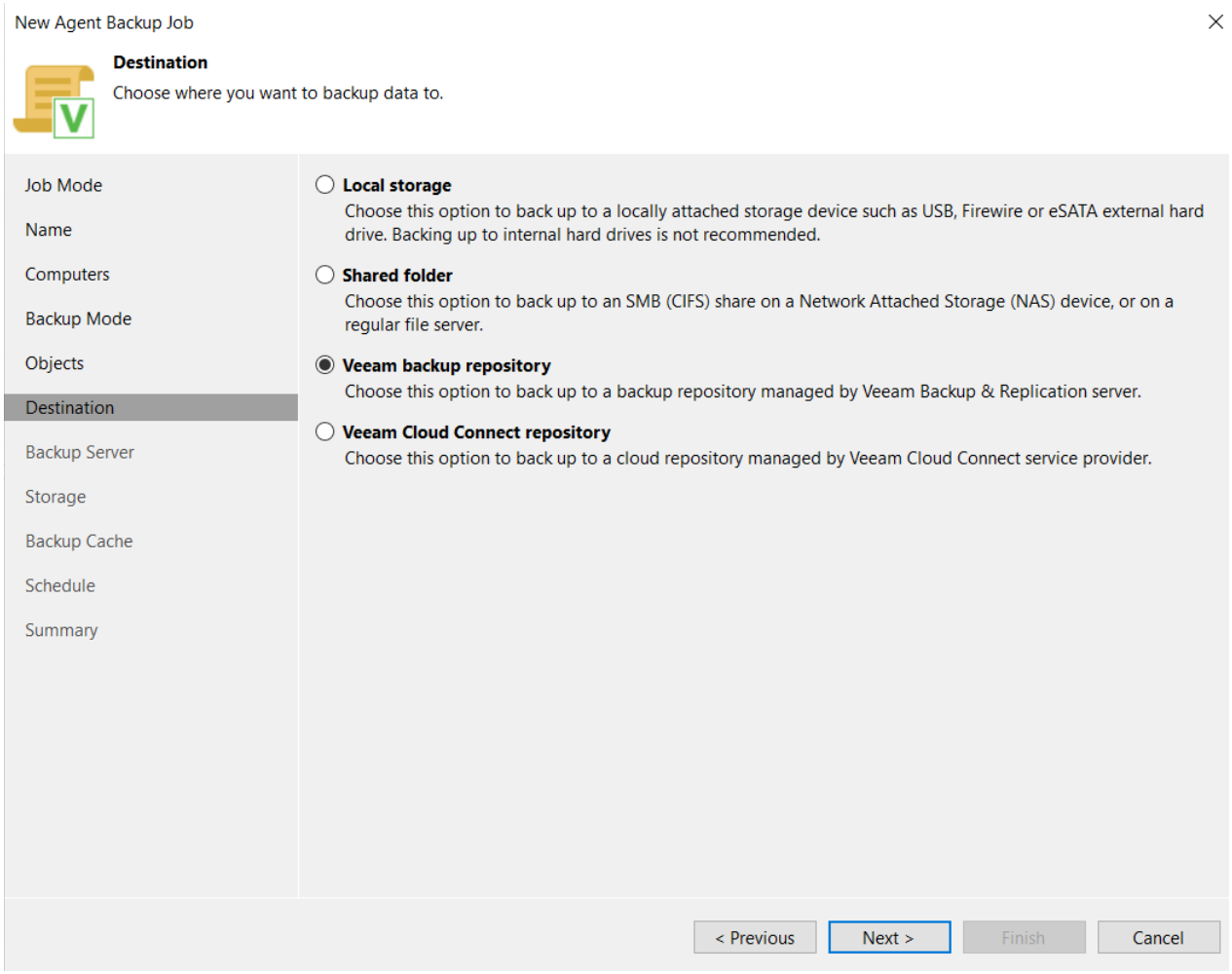


Figure 18. *Veeam backup repository selected as destination*

D-Backup-Repository was selected and a seven-day retention policy was configured.

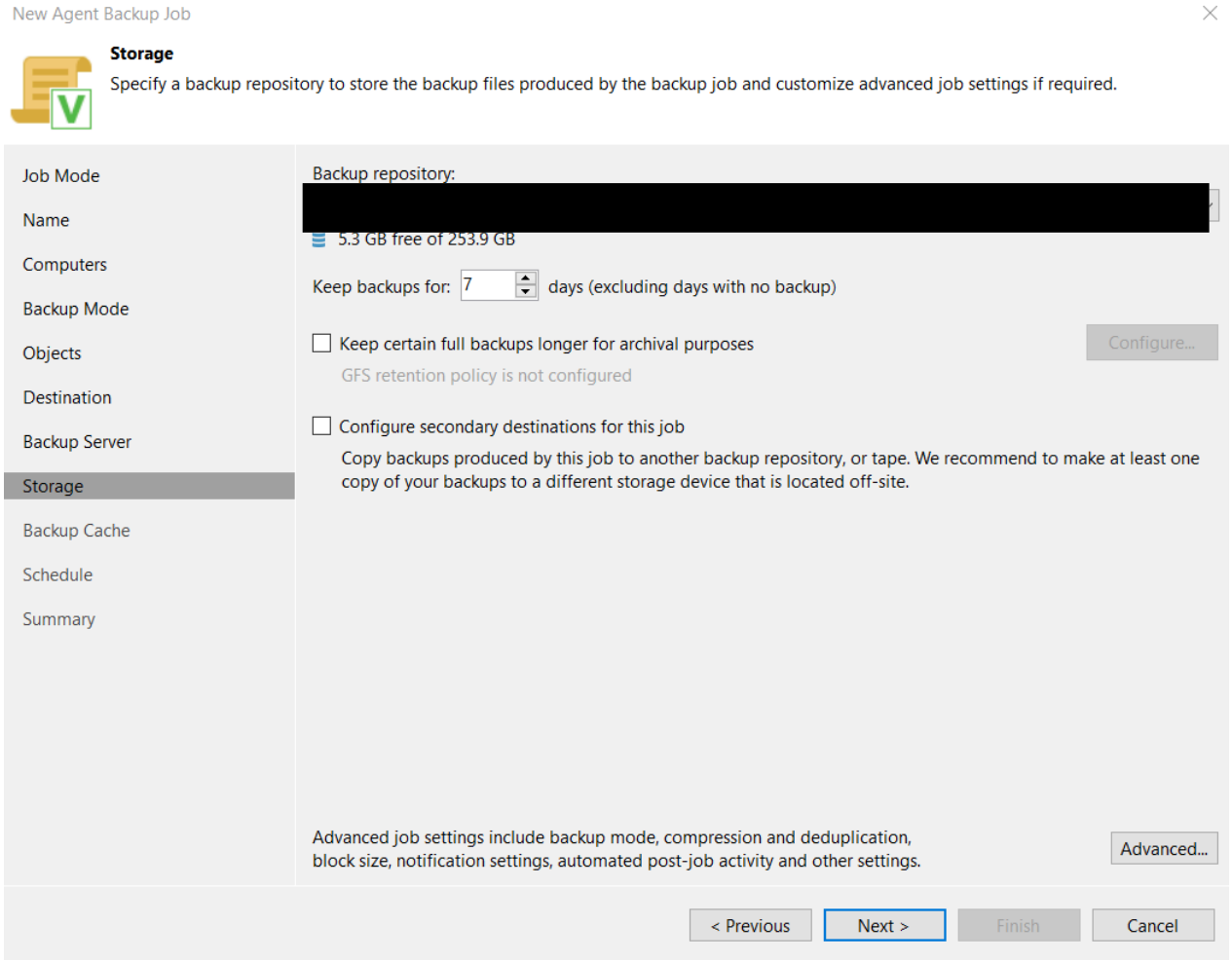


Figure 19. *Repository and retention configured*

The wizard summary confirmed workstation type, managed-by-agent mode, file-level backup, repository target, and manual execution.

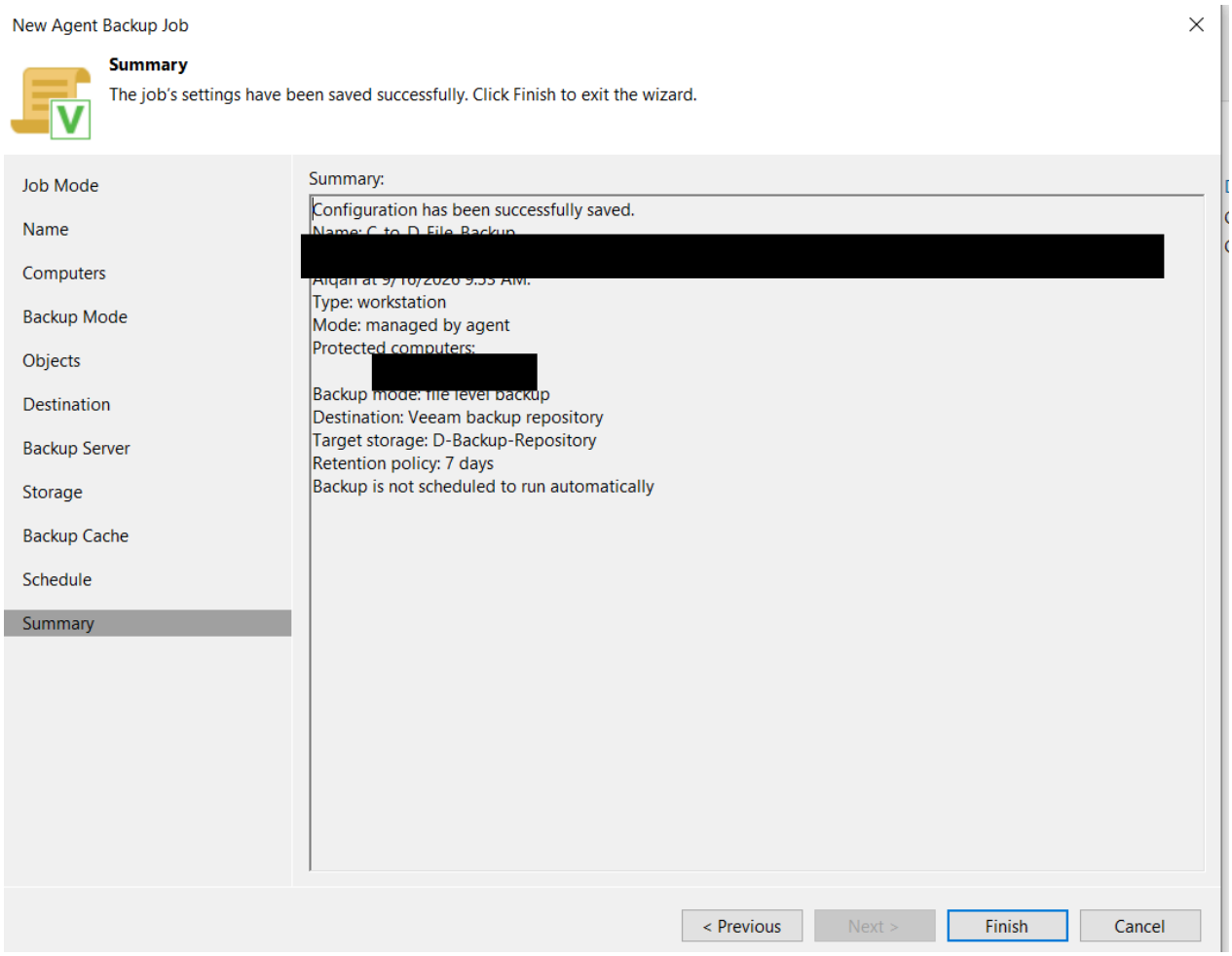


Figure 20. *Backup job summary reviewed*

3.4 Agent Access and Rescan Validation

After resolving access requirements, the protected computer was rescanned and processed successfully.

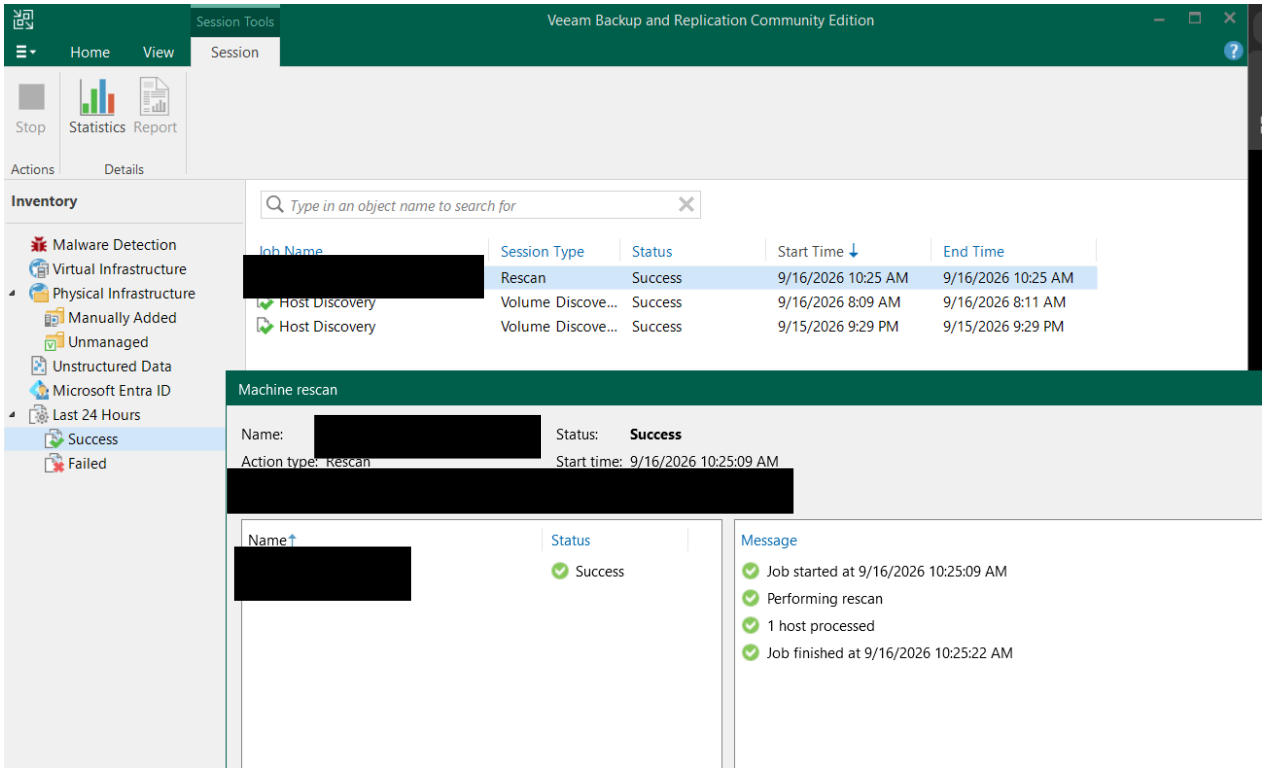


Figure 21. *Agent rescan completed successfully*

3.5 Backup Execution and Repository Verification

C_to_D_File_Backup appeared as an enabled Windows Agent Policy and was ready for manual execution.

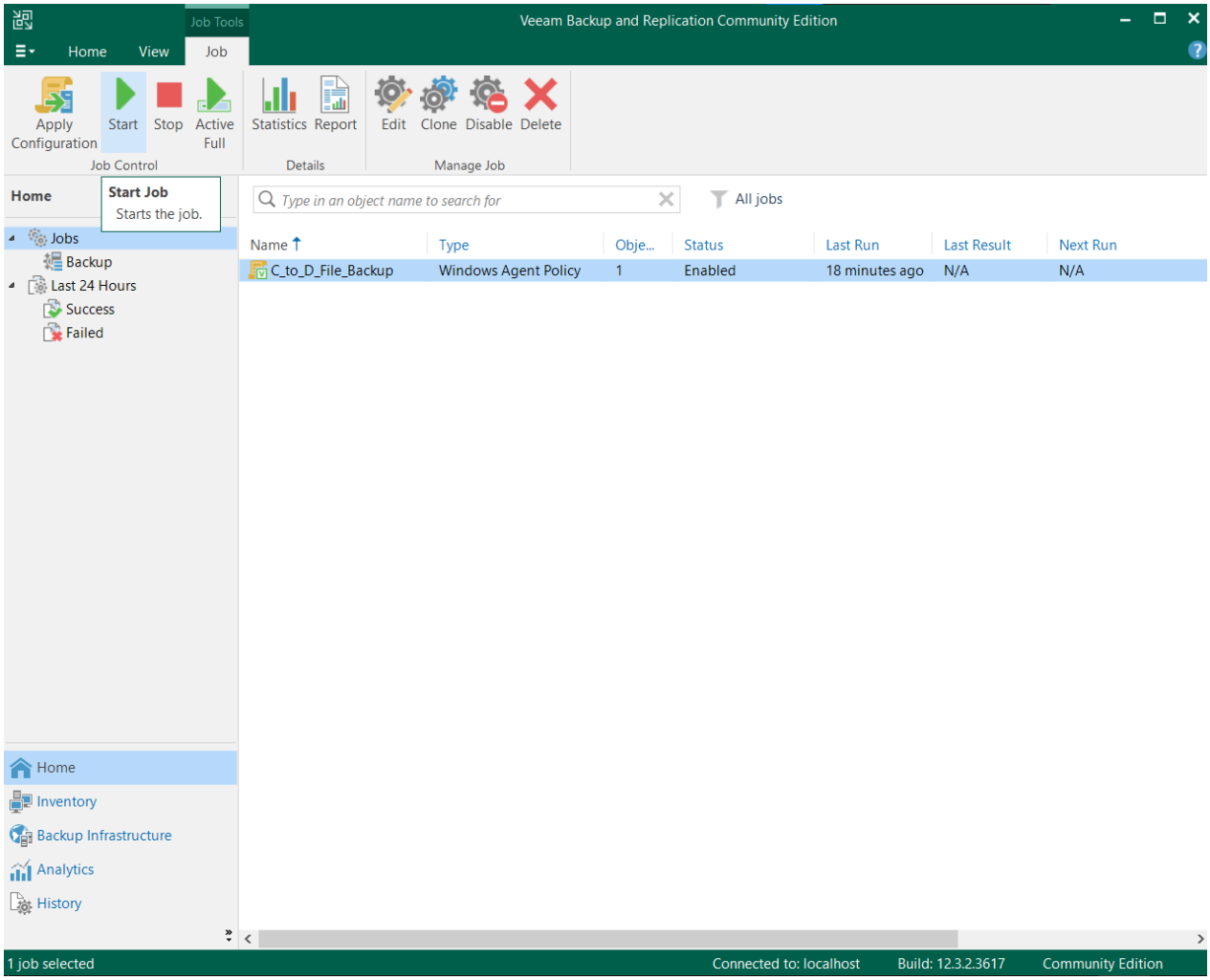


Figure 22. *Backup job ready to start*

The backup session started and was monitored from the Last 24 Hours view.

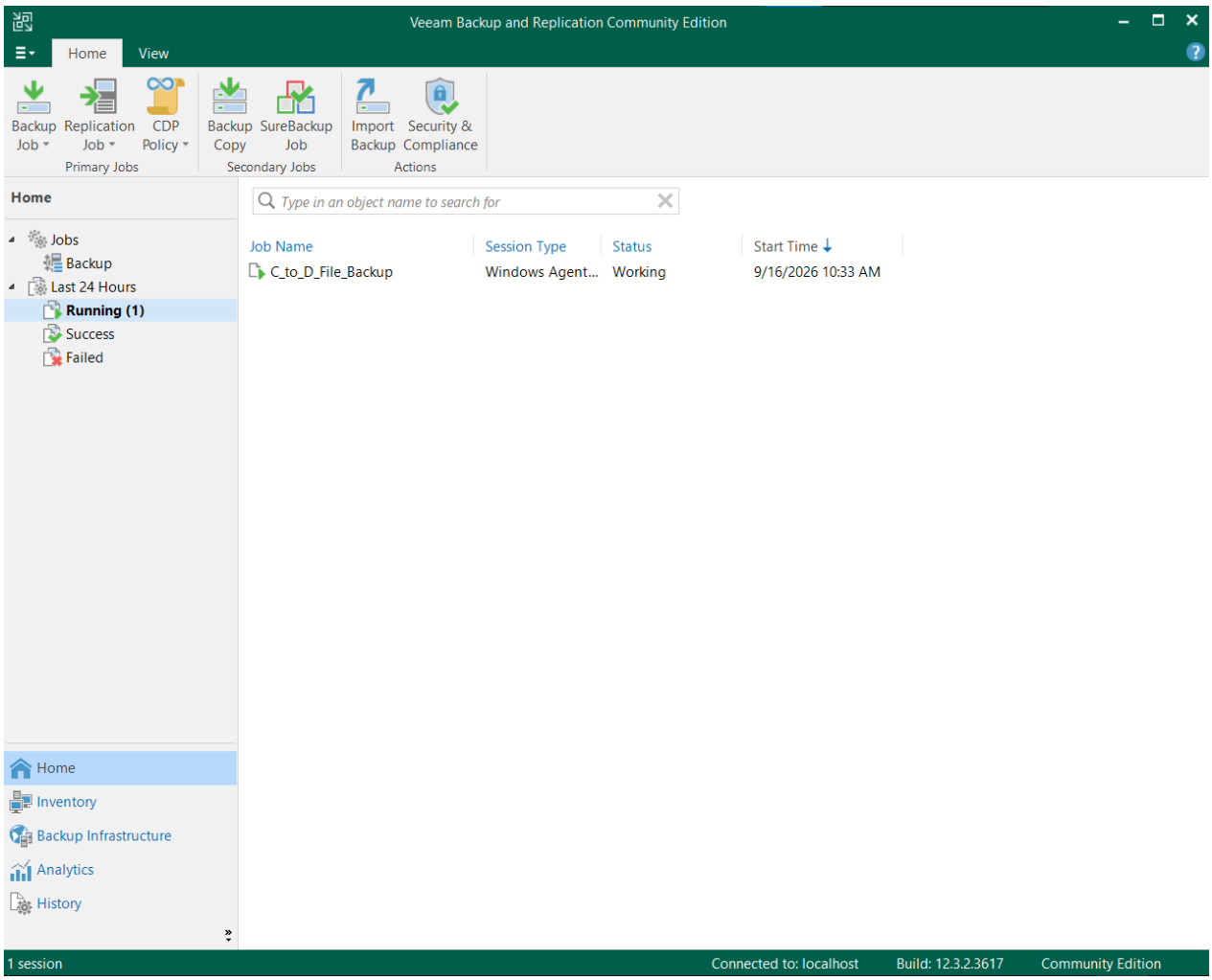


Figure 23. *Backup job running*

Veeam reported a successful Windows Agent backup session for the protected workstation.

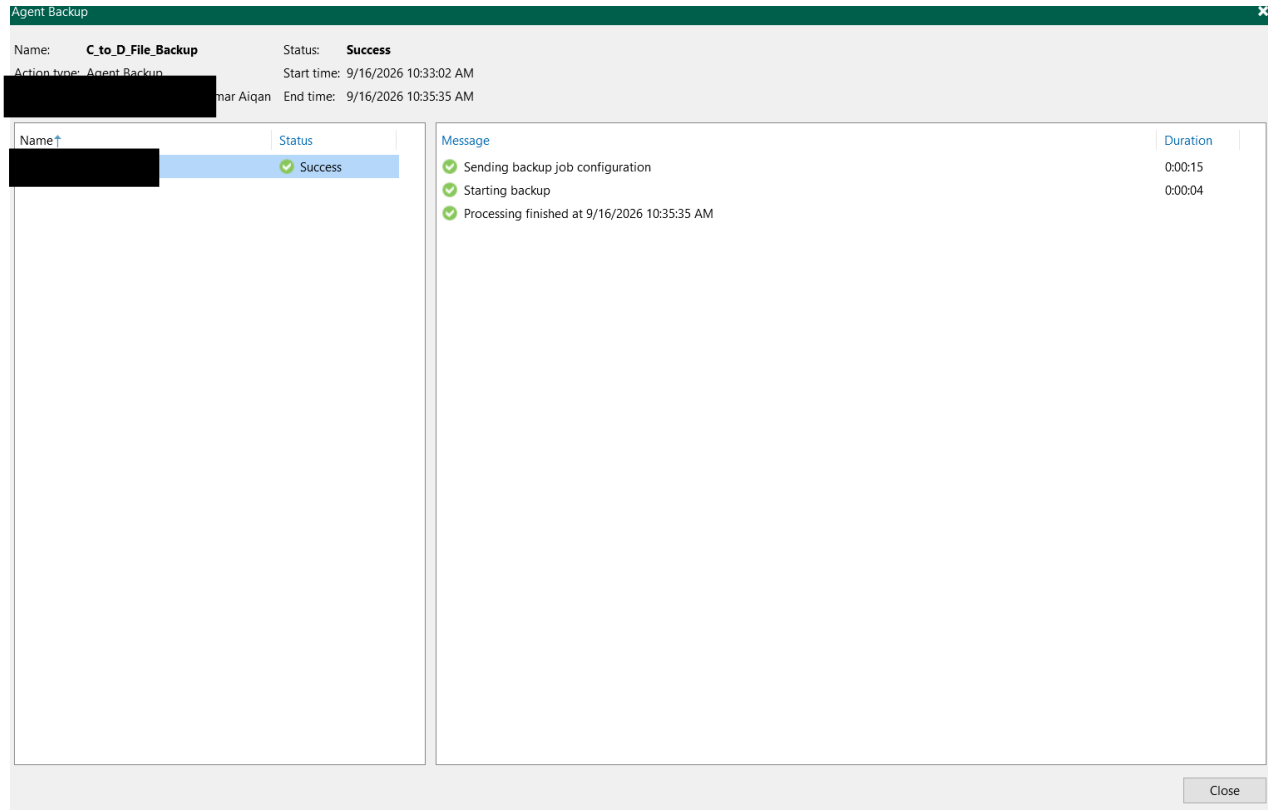


Figure 24. *Backup session completed successfully*

The repository contained Veeam backup metadata and backup data files (.vbm and .vbk), confirming physical creation of the backup set.

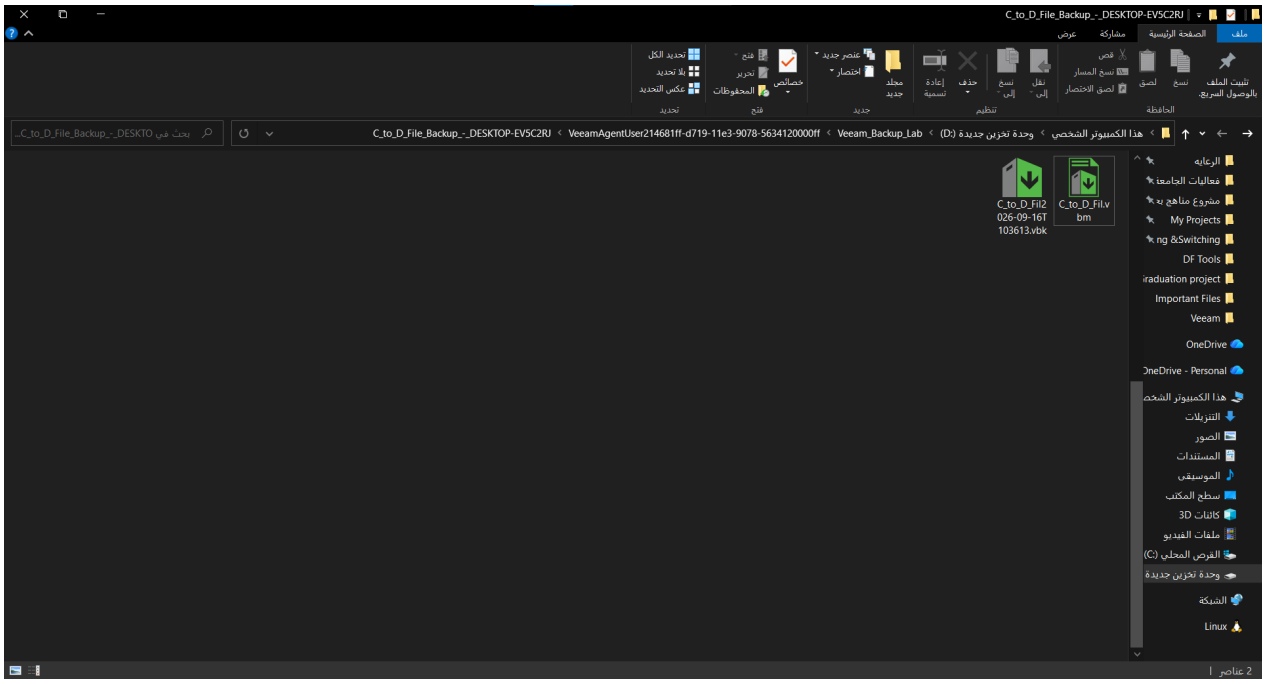


Figure 25. *Backup files created on drive D*

3.6 File Deletion and Recovery Validation

RestoreMe.txt was present in the source directory together with the other test files before the recovery test.

```
PS C:\Windows\system32> Get-ChildItem "C:\Veeam_Lab_Source"


    Directory: C:\Veeam_Lab_Source


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/16/2026   9:22 AM             27 File1.txt
-a----         9/16/2026   9:22 AM             34 Important.txt
-a----         9/16/2026   9:22 AM             19 RestoreMe.txt


PS C:\Windows\system32>
```

Figure 26. *Restore test file confirmed before deletion*

RestoreMe.txt was intentionally removed from the source directory to simulate accidental deletion.

```
PS C:\Windows\system32> Remove-Item "C:\Veeam_Lab_Source\RestoreMe.txt"
PS C:\Windows\system32> Get-ChildItem "C:\Veeam_Lab_Source"


    Directory: C:\Veeam_Lab_Source


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/16/2026   9:22 AM             27 File1.txt
-a----         9/16/2026   9:22 AM             34 Important.txt


PS C:\Windows\system32>
```

Figure 27. *Restore test file deleted*

The protected computer backup was selected from Backups > Disk and the file restore workflow was started.

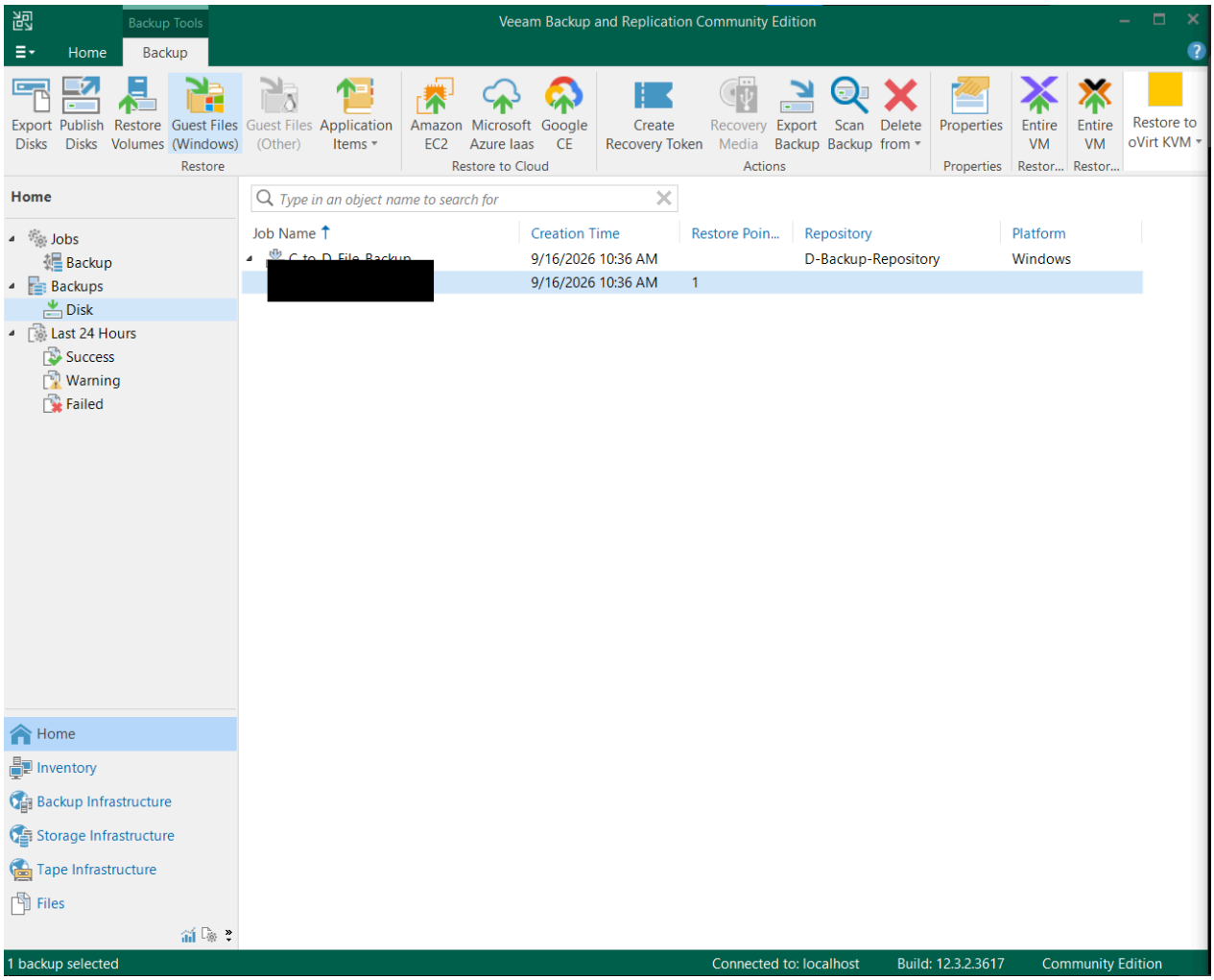


Figure 28. *Backup selected for file recovery*

The available restore point created by C_to_D_File_Backup was selected for the recovery operation.

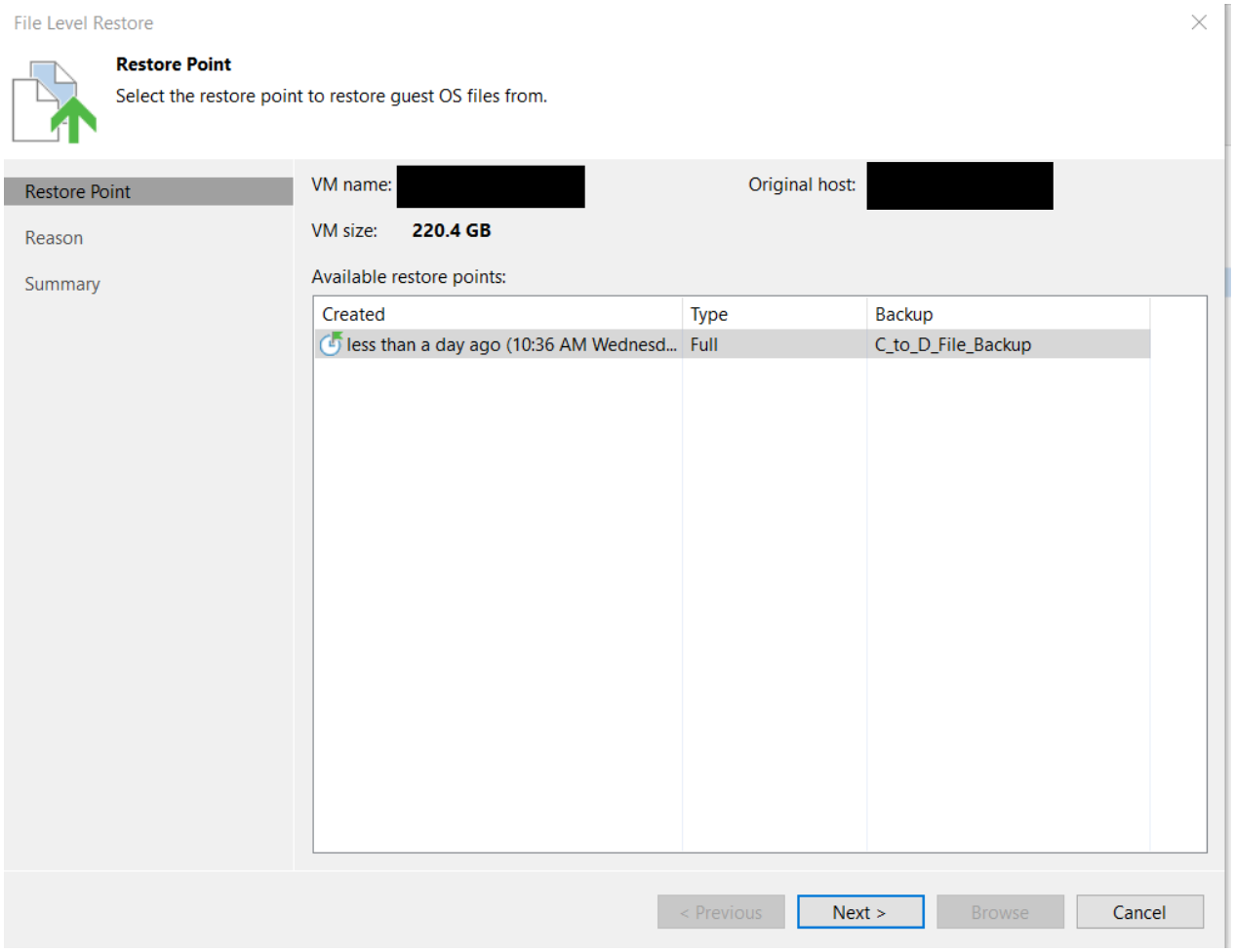


Figure 29. *Restore point selected*

The restore wizard summary confirmed the target computer and selected restore point before opening the backup browser.

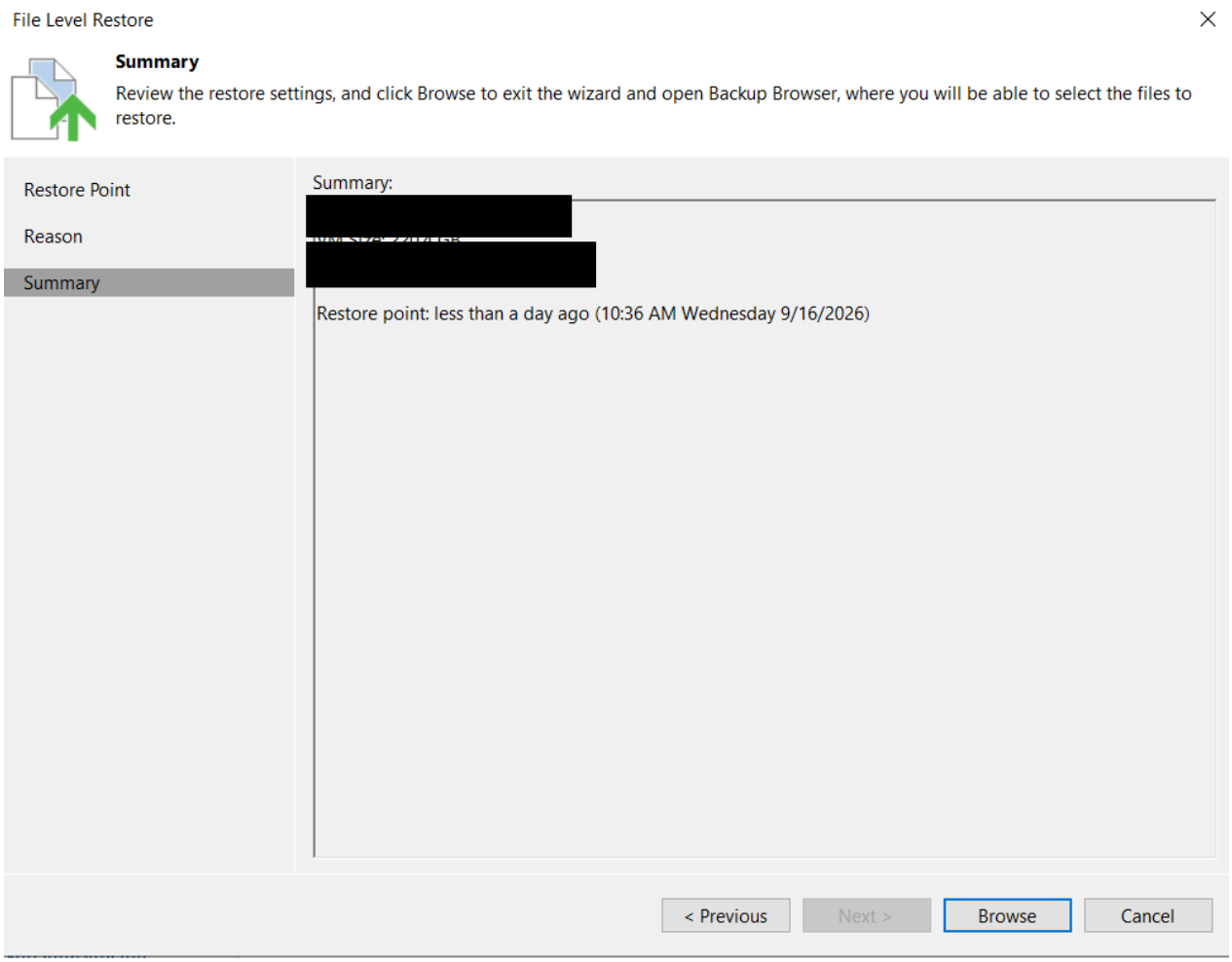


Figure 30. *File Level Restore summary*

Veeam started a Windows guest file restore session for the protected workstation.

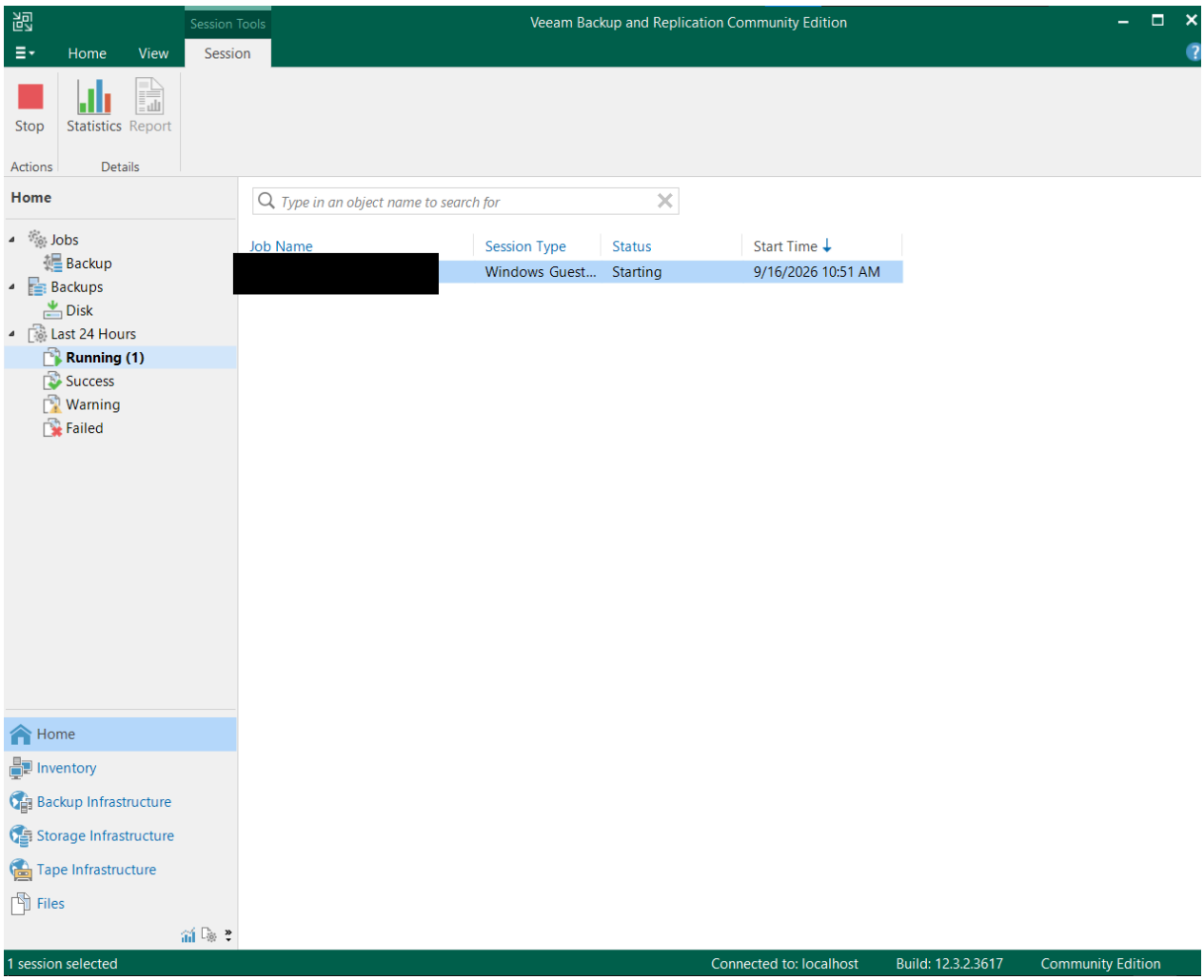


Figure 31. *File Level Restore session started*

Veeam Backup Browser showed RestoreMe.txt inside C:\Veeam_Lab_Source in the restore point.

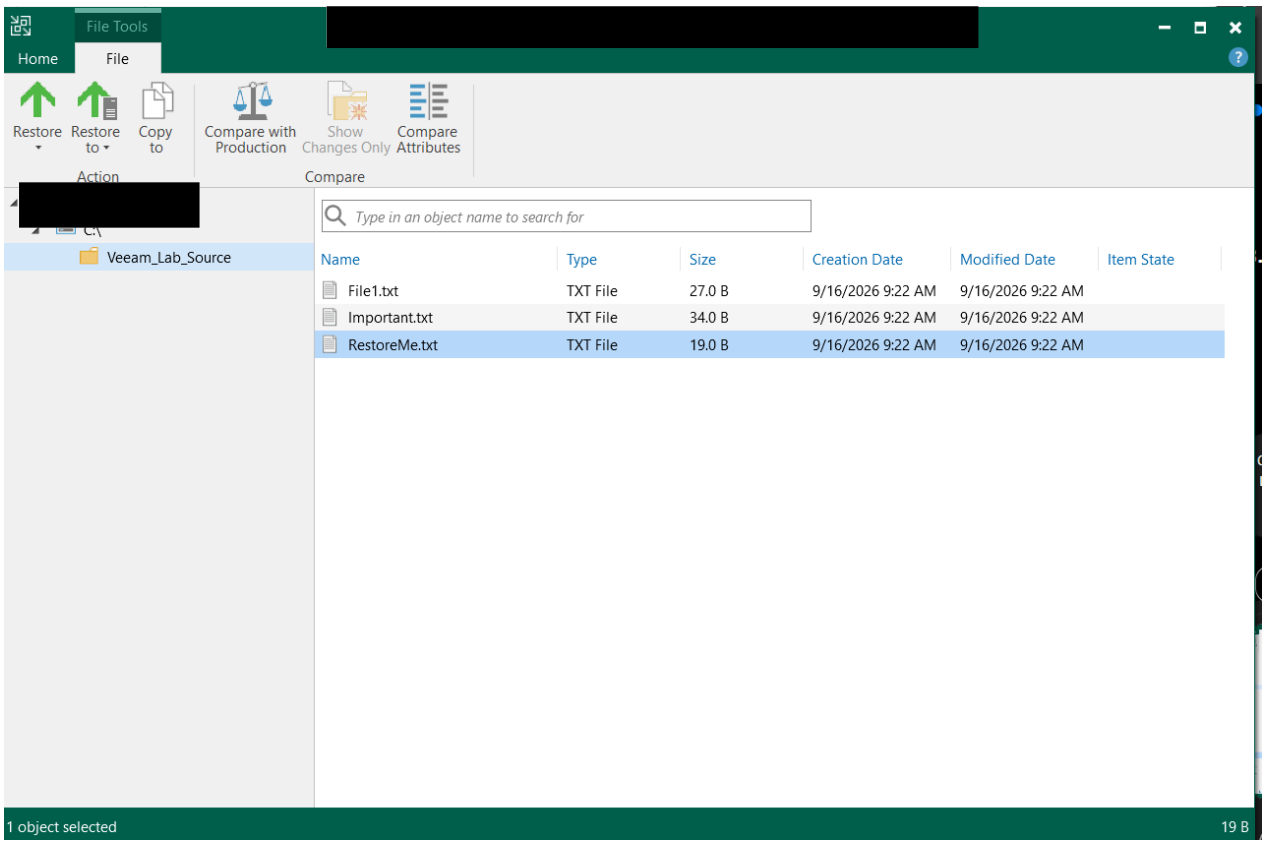


Figure 32. *Deleted file found inside the backup*

The restore progress window confirmed that RestoreMe.txt was restored successfully to the workstation.

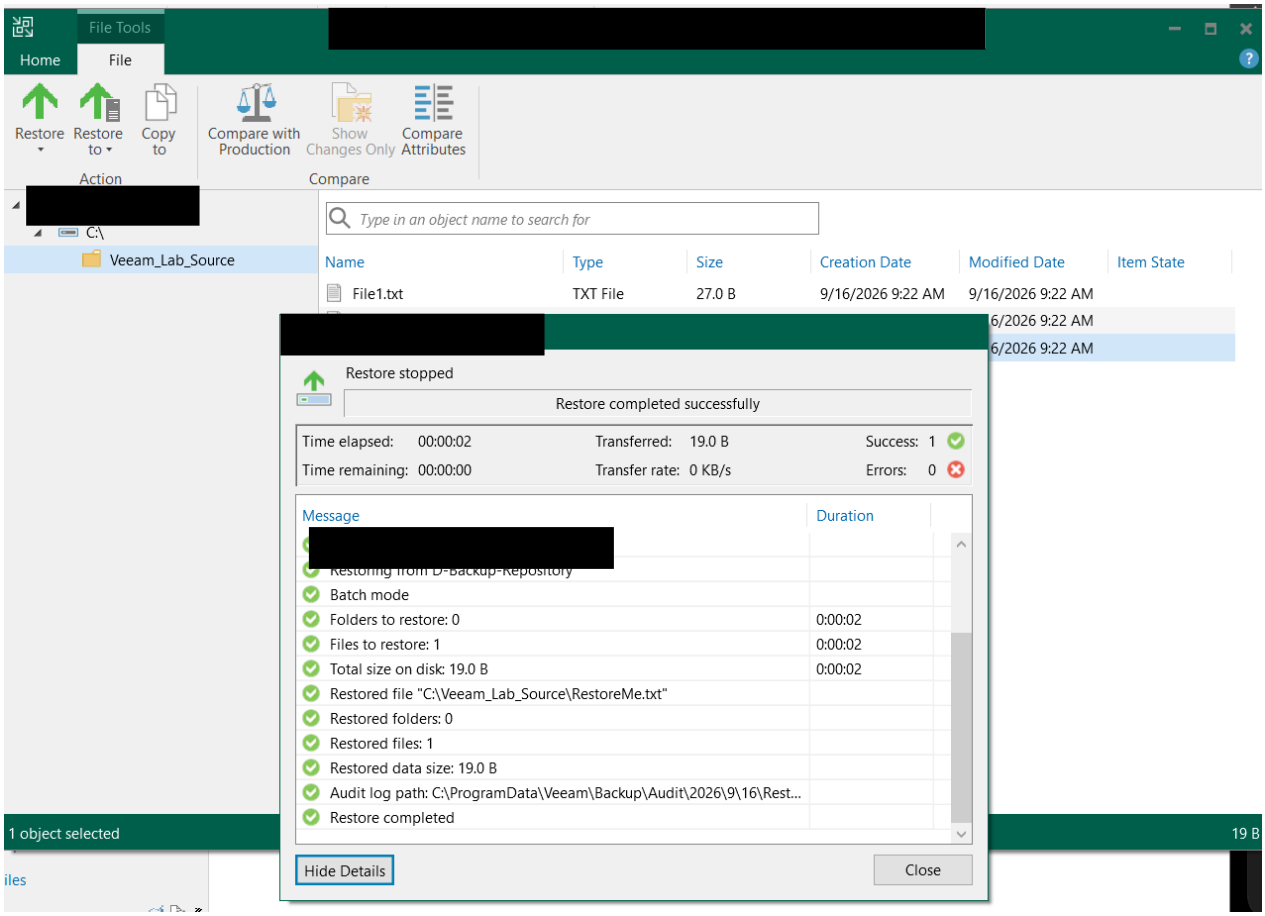


Figure 33. *File restored successfully*

PowerShell confirmed that RestoreMe.txt returned to C:\Veeam_Lab_Source and that its content was "Restore test file".

```
PS C:\Windows\system32> Get-ChildItem "C:\Veeam_Lab_Source"

    Directory: C:\Veeam_Lab_Source

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/16/2026   9:22 AM             27 File1.txt
-a----         9/16/2026   9:22 AM             34 Important.txt

PS C:\Windows\system32> Get-ChildItem "C:\Veeam_Lab_Source"

    Directory: C:\Veeam_Lab_Source

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         9/16/2026   9:22 AM             27 File1.txt
-a----         9/16/2026   9:22 AM             34 Important.txt
-a----         9/16/2026   9:22 AM             19 RestoreMe.txt

PS C:\Windows\system32> Get-Content "C:\Veeam_Lab_Source\RestoreMe.txt"
Restore test file
PS C:\Windows\system32> s
```

Figure 34. *Restored file verified in the source folder*

4 Troubleshooting and Security Considerations

Issues encountered, corrective actions, and operational lessons

4.1 Agent Deployment Access Denied

During the first rescan, Veeam could not deploy the backup agent because access to the Windows administrative share was denied. The corrective action was to use a dedicated temporary local administrator account and verify administrative-share connectivity. After the access configuration was corrected, the rescan completed successfully and one host was processed.

4.2 Administrative Share Validation

[redacted]

4.3 Low Repository Free Space

The repository had approximately 5.3 GB of free space remaining. Veeam can therefore complete this small laboratory backup, but the environment is close to its capacity threshold. A production repository should maintain substantial free capacity and should be monitored to prevent backup-chain failure or

[redacted]

service account with managed credentials.

5 Results and Verification

Evidence-based assessment of the completed lab

Control	Status	Evidence / Observation
Veeam installation	**PASS**	Community Edition launched successfully.
Repository creation	**PASS**	D-Backup-Repository created at D: \Veeam_Backup_Lab.
Backup policy configuration	**PASS**	Windows Agent file-level policy created for C:\Veeam_Lab_Source.
Agent connectivity	**PASS**	Protected computer rescan completed successfully.
Backup execution	**PASS**	Backup session completed and repository backup files were created.
Restore point availability	**PASS**	A valid restore point was available for file-level recovery.
Deleted-file recovery	**PASS**	RestoreMe.txt was recovered from the Veeam Backup Browser.
Post-restore verification	**PASS**	File presence and content were verified through PowerShell.
Repository capacity	**WARNING**	Approximately 5.3 GB free remained; additional capacity is recommended.

5.1 Recovery Verification Evidence

The strongest recovery evidence is the final PowerShell verification: the directory listing first showed only File1.txt and Important.txt after deletion, then showed RestoreMe.txt again after recovery. Get-Content returned the expected text "Restore test file", confirming content-level integrity of the restored object.

5.2 Overall Result

Overall laboratory result: SUCCESS. The configured Veeam workflow successfully protected selected data, created recoverable backup media, and restored an intentionally deleted file.

6 Conclusion

Final assessment

This practical exercise demonstrated the complete lifecycle of a Veeam file-level backup on Windows: repository preparation, policy creation, protected-host access, job execution, restore-point creation, file deletion, and file recovery. The backup repository contained the expected Veeam backup artifacts, and the recovery test restored the deleted file to its original path with the correct content.

The lab also highlighted two operational considerations that are important beyond the classroom: Veeam Agent deployment depends on valid administrative access to the protected host, and a backup repository must maintain adequate free capacity. These observations reinforce the need for proper credential management, storage monitoring, and post-lab security cleanup.

BACKUP CREATED ✓ RESTORE COMPLETED ✓ CONTENT VERIFIED ✓

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